



MIPI CSI-2 to USB UVC Reference Design User Guide

Reference Design

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Abbreviations in This Document

A list of abbreviations used in this document.

Abbreviations	Definition
AHB-Lite	Advanced High-Performance Bus – Lite
APB	Advanced Peripheral Bus
AXI4	Advanced eXtensible Interface 4
BPC	Bits per Component
CCM	Color Correction Matrix
CPU	Central Processing Unit
CSC	Color Space Converter
CSI-2	Camera Serial Interface-2
DMA	Direct Memory Access
DSI	Display Serial Interface
EBR	Embedded Block RAM
FIFO	First In First Out
FPGA	Field Programmable Gate Array
FPS	Frames per Second
GPIO	General Purpose Input Output
GUI	Graphical User Interface
HDL	Hardware Description Language
IEBM	In Endpoint Buffer Manager
I2C	Inter-Integrated Circuit
IP	Intellectual Property
IPK	IP Package
ISP	Image Signal Processing
LMMI	Lattice Memory Mapped Interface
MIPI	Mobile Industry Processor Interface
PC	Personal Computer
PHY	Physical Layer
PLL	Phase-Locked Loop
PPC	Pixels per Clock
RAM	Random Access Memory
RGB	Red, Green, Blue
RISC-V	Reduced Instruction Set Computer – Five (V)
RISC-V MC	RISC-V for Microcontroller Applications
RTL	Register Transfer Level
SCL	I2C controller serial clock line
SDA	I2C controller serial data line
SDK	Software Development Kit
SPI	Serial Peripheral Interface
TDP	True Dual-Port
TPG	Test Pattern Generator
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
UVC	USB Video Class
UVSI	Unified Video Streaming Interface

1. Introduction

The Lattice Semiconductor CrossLinkU™-NX Mobile Industry Processor Interface (MIPI®) Camera Serial Interface-2 (CSI-2) to Universal Serial Bus (USB) Video Class (UVC) reference design provides a template for unified video streaming from a camera sensor, using the USB hard IP in a CrossLinkU-NX device.

1.1. Quick Facts

Download the design files for the MIPI CSI-2 to USB UVC Reference Design from the corresponding [GitHub](#) page.

Table 1.1. Summary of the Reference Design

General	Target Devices	LIFCL-33U
	Source code format	C Code, Verilog and SystemVerilog RTL
Simulation	Functional simulation	Not performed
	Timing simulation	Not performed
	Test bench	Not available
	Test bench format	Not applicable
Software Requirements	Software tool and version	Lattice Propel™ SDK 2026.1 Lattice Propel Builder 2026.1 Lattice Radiant™ Software 2026.1 Lattice Radiant Programmer 2026.1 PotPlayer Software (3 rd Party), version 1.7 – Download from Microsoft Store or PotPlayer Official Website
	IP version ¹ (if applicable)	MIPI CSI/DSI IP Core version 4.2.0 Color Space Converter IP Core version 2.4.1.03 (Custom IP) Color Correction Matrix IP Core version 1.3.1.03 (Custom IP) Debayer IP Core version 1.4.0.03 (Custom IP) USB23 IP Core version 3.2.0
Hardware Requirements	Board	CrossLinkU-NX Evaluation Board REV-B (LIFCL-33U-EVN)
	Camera	Raspberry Pi Camera Module V2
	Cable	USB Type-C to USB Type-C cable, or USB Type-C to USB Type-A 9-pin cable USB Type-A to Micro USB cable

Note:

1. In some instances, the IPs may be updated without changes to the corresponding user guides. The user guide of each IP may reflect an earlier IP version but remains fully compatible with the later IP version.

1.2. Features

Key features of the MIPI CSI-2 to USB UVC reference design include:

- The Lattice MIPI CSI/DSI IP Core in this reference design is configured to support two lanes (capped by IMX219 capability) with a lane rate of 592 Mbps (capped by driver configuration). The IP receives, processes, and converts incoming MIPI video payload packets into Unified Video Streaming Interface (UVSI) packets.
- The Lattice Debayer IP Core extracts the R, G, and B components from the pixel data output by the image sensor, converting the RAW10 data format into full RGB components. For details, refer to [<IP Installation Path>/latticesemi_custom_ip_debayer_1.4.0.03/doc/IP_Design_Notes.pdf](#).
- The Lattice Color Correction Matrix IP Core performs pixel data correction by adjusting R, G, and B components gain and compensates for color channel crosstalk. For details, refer to [<IP Installation Path>/latticesemi_custom_ip_ccm_1.3.1.03/doc/IP_Design_Notes.pdf](#).
- The Color Space Converter IP Core converts RGB data into the YUV format. For details, refer to [<IP Installation Path>/latticesemi_custom_ip_color_space_converter_2.4.1.03/doc/IP_Design_Notes.pdf](#).
- The Chroma Resampler converts YUV444 data into the YUV422 format by reducing the chroma component.

- The Video to USB Bridge takes in UVSI packets and forms a data stream in USB Video Class (UVC) format for USB transfer.
- The hardware USB IP Core streams the unified video data from the image sensor to a PC through a USB Type-C connector.

1.3. Naming Conventions

1.3.1. Nomenclature

The nomenclature used in this document is based on Verilog HDL and SystemVerilog HDL.

1.3.2. Signal Names

- `_n` are active low, asserted when value is logic 0.
- `_i` are input signals.
- `_o` are output signals.
- `_io` are bi-directional input and output signals.

2. Directory Structure and Files

Figure 2.1 shows the directory structure.

```

+---docs //Contains reference design user guide
+---fpga_lifcl //FPGA_$DEVICE_FAMILY
|   +---precompiled_file //Contains a precompiled bitstream (IP Evaluation) for the default configuration
|   +---propelsdk
|   |   +---riscv_mc
|   \---radiant //Contains Lattice Propel Builder files, Radiant files and RTL source code
|       +---propelbld_mipi_csi_to_usb_uvc
|       |   \---lib
|       |       +---gluelogics
|       |       +---IPs
|       |       \---latticesemi.com
|       |           +---ip
|       |           |   +---ahbl_to_axilite_bridge
|       |           |   +---ahbl_to_lmml_inst
|       |           |   +---ahbl_to_mem_inst
|       |           |   +---cpu0
|       |           |   +---HW_ver_gpio
|       |           |   +---lsc_i2cc
|       |           |   +---mipi_csi_rx
|       |           |   +---system0
|       |           |   +---tp_en_gpio
|       |           |   +---uart0
|       |           |   +---unified_interconnect_0
|       |           \---module
|       |               +---pll_multiclk
|       |               +---rs_pixel
|       |               +---rs_riscv
|       |               +---rs_usb
|       |               \---USB23
|       \---latticesemi_custom
|           +---ip
|           |   +---isp_ccm
|           |   +---isp_chroma_resampler
|           |   +---isp_csc
|           |   +---isp_debayer
|           |   +---unified_video_to_usb
|           |   +---usb_bus_interface_bridge
|           \---sge //System Generation Environment (Generated by Propel Builder)
|       +---src
|       |   \---constraints //Contains the constraints file (.pdc) used in the design
|       \---verification //Generated by Propel Builder
+---misc //Contains non-FPGA files
|   +---custom_ip //Contains Lattice IP Packagers for custom IPs (eg: ISP IP, AHBL to AXI4-Lite, and etc)

```

Figure 2.1. Directory Structure

Table 2.1 shows the list of files included in the reference design package.

Table 2.1. File List

Attribute	Description
<Component name>.ipx	This file contains the information on the files associated with the generated IP.
<Component name>.cfg	This file contains the parameter values used in IP configuration.
component.xml	This file contains the ipxact:component information of the IP.
design.xml	This file documents the configuration parameters of the IP in IP-XACT 2014 format.
rtl/<Component name>.v	This file provides an example RTL top file that instantiates the module.
rtl/<Component name>_bb.v	This file provides the synthesis closed-box.
misc/<Component name>_tmpl.v misc/<Component name>_tmpl.vhd	These files provide instance templates for the module.

3. Functional Description

The top-level block diagram of the CrossLinkU-NX MIPI CSI-2 to USB UVC reference design is shown in Figure 3.1 below.

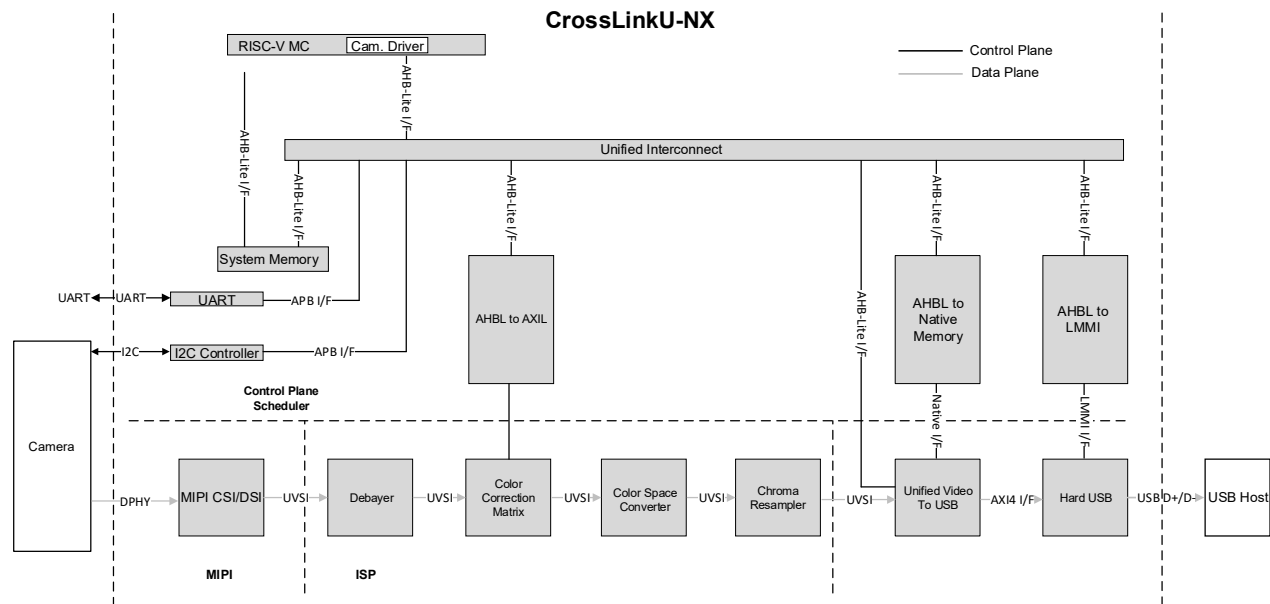


Figure 3.1. Reference Design Block Diagram

3.1. Design Components

The CrossLinkU-NX MIPI CSI-2 to USB UVC reference design includes blocks described in the following sections. In this reference design, video data is captured through the CSI-2 interface using the RAW10 color format. The incoming RAW10 data stream is converted to RGB888, subsequently YUV444, and finally YUV422 through the image signal processing (ISP) pipeline. The ISPs operate with a configuration of 8 bits per component (BPC) and 2 pixels per clock (PPC), with a maximum supported video resolution of 3280x2160.

3.1.1. USB23 Controller

This is the USB23 Device Controller hardened IP. This IP communicates with the system memory through the Advanced eXtensible Interface 4 (AXI4) to Advanced High-Performance Bus – Lite (AHB-Lite) converter module.

The AXI4 Direct Memory Access (DMA) Controller of the USB23 Controller is connected to two different memories through the AXI4 Interconnect Bridge.

- For Bulk transfer, the USB controller fetches data from the In Endpoint Buffer Manager (IEBM) module's data Random Access Memory (RAM). This RAM is not shared with any other controller or RISC-V processor.
- For all other operations, the USB controller talks to the True Dual-Port (TDP) Memory, which is shared with the RISC-V processor. This kind of architecture helps to achieve higher Bulk throughput.

3.1.2. AHB-Lite to LMMI Converter

The USB23 controller register interface can be configured through its Lattice Memory Mapped Interface (LMMI). Since RISC-V MC supports only the AHB-Lite interface, an AHB-Lite to LMMI converter is needed.

3.1.3. RISC-V Microcontroller

All the serial peripheral blocks are connected to the RISC-V MC soft processor through the AHB-Lite interface. The RISC-V MC CPU IP processes data and instructions while monitoring external interrupts coming from the USB23 controller and IEBM core.

3.1.4. System Memory

This IP is a soft IP for memory which is used by the RISC-V MC processor. The processor reads the instructions stored in this memory through the AHB-Lite interface and takes actions accordingly. This IP has two interfaces, AHB-Lite and AXI4, through which the manager can communicate with it. The AHB-Lite interface is used to communicate with the RISC-V processor.

3.1.5. UART

The RISC-V MC processor can communicate with the UART and GPIO cores through its Advanced Peripheral Bus (APB) interface. Hence, the UART and GPIO cores are connected to the soft processor through the AHB-Lite to APB Bridge. The UART IP performs serial-to-parallel conversion on data characters received from a peripheral UART device. The IP performs parallel-to-serial conversion on data characters received from the host through an APB interface.

3.1.6. GPIO

There is a 32-bit input GPIO, which is the hardware version GPIO, to monitor the FPGA design version for a particular release. Also, there is a 2-bit output GPIO, known as the test pattern enable GPIO. This GPIO is used to select between the Image Signal Processing data stream and YUV422 Test Pattern Generator.

3.1.7. Lattice I2C Controller

This Lattice I2C Controller IP is designed for communicating with any I2C target device. It supports standard operations such as I2C write, I2C read, and combined write-then-read transactions with the connected device. In this design, the RISC-V MC processor uses the I2C Controller IP to configure the registers of the MIPI camera sensor, enabling proper initialization and operation of the camera.

3.1.8. Unified Interconnect

This IP provides the communication interface between the RISC-V soft processor and peripherals that support AHB-Lite and APB interfaces, such as System Memory, UART, and I2C Controller.

3.1.9. MIPI CSI/DSI IP Core

The Lattice MIPI CSI/DSI IP provides a flexible solution for receiving high-speed video data over a MIPI-compliant D-PHY interface in FPGA designs. MIPI enables high-speed, low-power video transmission between components such as cameras and displays. In this reference design, the video packets are received through the D-PHY physical layer interface from the Raspberry Pi Camera Module V2. The incoming video payload packets are processed and converted from bytes to pixels before being transmitted through UVS1.

3.1.10. Debayer IP Core

The Debayer IP accepts pixel data input in Bayer format from a source such as an image sensor for processing. To extract the individual R, G, and B color components from this Bayer-patterned data, a Debayer function is required. The Debayer IP converts RAW10 image data into full RGB components. This IP supports UCSI interfaces for both its input and output video streams. For complete specifications, including interface signals and parameter details, refer to the Debayer IP documentation:

<IP Installation Path>/latticesemi_custom_ip_debayer_1.4.0.03/doc/IP_Design_Notes.pdf.

3.1.11. Color Correction Matrix IP Core

To obtain the correct colors, the IP performs pixel data correction by adjusting R, G, and B components gain and compensating for color channel crosstalk. This linear mapping of the color components is achieved by using a 3 x 3 matrix, called color correction matrix (CCM). For complete specifications, including interface signals and parameter details, refer to the Color Correction Matrix IP documentation:

<IP Installation Path>/latticesemi_custom_ip_ccm_1.3.1.03/doc/IP_Design_Notes.pdf.

3.1.12. Color Space Converter IP Core

The Color Space Converter IP block converts RGB data into the YUV format. For complete specifications, including interface signals and parameter details, refer to the Color Space Converter IP documentation:

<IP Installation

Path>/latticesemi_custom_ip_color_space_converter_2.4.1.03/doc/IP_Design_Notes.pdf.

3.1.13. Chroma Resampler

The Chroma Resampler converts YUV444 data into the YUV422 format by reducing the chroma component.

3.1.14. Video to USB Bridge

The Video to USB Bridge is responsible for receiving video data from UCSI, packing into a UVC-compliant format, buffering using In Endpoint Buffer Manager (IEBM) and True Dual-Port (TDP) Memory, and handling the transfer between the data buffer and the USB23 controller for host transmission. In addition, the Video to USB Bridge features a YUV422 Test Pattern Generator (TPG) for debugging available for use by enabling it using the control plane. This TPG only supports 1080p resolution.

To support the USB data flow, the bridge uses an IEBM to organize and store video data before it reaches the USB23 controller. The IEBM writes video data sequentially into endpoint buffers through a FIFO interface and then exposes these buffers to the USB23 controller through AXI4 for reading. This ensures continuous and orderly movement of video frames through the system. The TDP Memory offers dedicated storage that supports simultaneous read and write operations between RISC-V CPU and USB23 Controller, helping maintain stable video throughput even at high data rates.

This module enables seamless data flow from UCSI to the USB controller while being easy to use, minimizing design blocks required to be instantiated by the user to achieve the transfer.

Refer to [Table 3.1](#) and [Table 3.2](#) for parameter and signal descriptions of this module.

Table 3.1. Video to USB Bridge Parameters

Port Name	Selectable Values	Default	Description
COLOR_FORMAT	YUV422	YUV422	Color Format
BPC	8	8	Bits Per Component
PIXEL_PER_CLK	1, 2, or 4	2	Pixels Per Clock

Table 3.2. Video to USB Bridge Signals

Port Name	I/O	Width	Description
pixel_clk	Input	1	Pixel Clock
clk_pll_72m	Input	1	72 MHz Clock generated by PLL

Port Name	I/O	Width	Description
mipi_byte_clk	Input	1	MIPI Byte Clock
system_reset_n	Input	1	AHB-Lite reset (Active Low)
pixel_clk_reset_n	Input	1	Reset for logic operating in the pixel clock domain (Active Low)
mipi_byte_clk_resetrn	Input	1	Reset for MIPI byte-clock domain logic (Active Low)
yuy2_tp_en	Input	1	Enables the internal YUY2 (YUV422) test-pattern generator through GPIO.
Native Memory Interface for True Dual-Port Memory			
mem_to_ahbl_32_wr_en_b	Input	1	Write-enable signal for writing data
mem_to_ahbl_32_wr_data_b	Input	[13:0]	Write-data bus for buffered memory writes
mem_to_ahbl_32_ben_b	Input	[3:0]	Byte-enable signals for selecting active write bytes
mem_to_ahbl_32_addr_b	Input	[9:0]	Address bus for memory transactions
mem_to_ahbl_32_rd_data_b	Output	[31:0]	Read-data output
AHB-Lite Subordinate Interface for IEBM			
ahbl_haddr_i	Input	[7:0]	AHB-Lite address bus for subordinate access.
ahbl_hburst_i	Input	[2:0]	Burst-type indicator for AHB-Lite transactions.
ahbl_hmastlock_i	Input	1	Manager-lock signal indicating locked AHB sequences
ahbl_hprot_i	Input	[3:0]	Protection control signals for privilege/cache/buffer settings
ahbl_hready_i	Input	1	Indicates that the previous transfer is complete. It allows new transfers.
ahbl_hsel_i	Input	1	AHB-Lite subordinate select signal
ahbl_hsize_i	Input	[2:0]	Specifies the transfer size, byte, half-word, or word.
ahbl_htrans_i	Input	[1:0]	AHB-Lite transfer type
ahbl_hwdata_i	Input	[31:0]	Write data from AHB-Lite manager
ahbl_hwrite_i	Input	1	Write direction indicator. 1=WRITE, 0=READ.
ahbl_hrdata_o	Output	[31:0]	Read data returned to AHB-Lite manager
ahbl_hreadyout_o	Output	1	Indicates the subordinate is ready to complete the current transfer.
ahbl_hresp_o	Output	1	Transfer response
AXI4 Subordinate Interface for USB23 Controller			
u23_axim_XMAWADDR	Input	[31:0]	AXI4 write-address channel: write address
u23_axim_XMAWBURST	Input	[1:0]	Burst type for write transactions
u23_axim_XMAWID	Input	[7:0]	Write-transaction ID
u23_axim_XMAWLEN	Input	[7:0]	Number of beats in a write burst
u23_axim_XMAWPROT	Input	[2:0]	Protection bits for write transactions
u23_axim_XMAWSIZE	Input	1	Write data size per beat (encoded)
u23_axim_XMAWREADY	Output	1	Indicates acceptance of write address.
u23_axim_XMAWVALID	Input	1	Write-address valid signal
u23_axim_XMWDATA	Input	[63:0]	Write-data bus
u23_axim_XMWSTRB	Input	[7:0]	Write strobe bits per byte
u23_axim_XMWLAST	Input	1	Indicates last beat of write burst
u23_axim_XMWREADY	Output	1	Indicates acceptance of write data
u23_axim_XMWVALID	Input	1	Write-data valid signal
u23_axim_XMARADDR	Input	[31:0]	AXI4 read-address channel: read address
u23_axim_XMARBURST	Input	[1:0]	Burst type for read transactions
u23_axim_XMARID	Input	[7:0]	Read-transaction ID
u23_axim_XMARLEN	Input	[7:0]	Number of beats in a read burst
u23_axim_XMARPROT	Input	[2:0]	Protection bits for read transactions

Port Name	I/O	Width	Description
u23_axim_XMARSIZE	Input	[2:0]	Read data size per beat
u23_axim_XMARREADY	Output	1	Indicates acceptance of read address.
u23_axim_XMARVALID	Input	1	Read-address valid signal
u23_axim_XMRDATA	Output	[63:0]	Read-data bus
u23_axim_XMRID	Output	[7:0]	ID of the read-data response
u23_axim_XMRRESP	Output	[1:0]	Read response type
u23_axim_XMRLAST	Output	1	Indicates the last beat of a read burst.
u23_axim_XMRREADY	Input	1	Indicates the Manager is ready to accept read data.
u23_axim_XMRVALID	Output	1	Read-data valid signal
u23_axim_XMBID	Output	[7:0]	ID for write-response channel
u23_axim_XMBRESP	Output	[1:0]	Write response
u23_axim_XMBREADY	Input	1	Indicates the Manager is ready for write response.
u23_axim_XMBVALID	Output	1	Write-response valid
Unified Video Streaming Interface			
tdata	Input	[PIXEL_PER_CLK*16-1:0]	TDATA is the primary payload that is used to provide the data that is passing across the interface.
tuser	Input	[1:0]	Sideband information is transmitted alongside the data stream. tuser[0] – start of frame tuser[1] – interlaced video
tlast	Input	1	TLAST indicates the end of the line for each horizontal line. It always coincides with the last pixel of the line.
tready	Output	1	TREADY indicates that RX can accept a transfer in the current cycle.
tvalid	Input	1	TVALID indicates availability of valid pixel data on tx_tdata_o when TREADY is asserted.
Interrupt			
iebm_ri_irq_o	Output	1	Interrupt request generated by IEBM to RISC-V indicating buffer ready to read.

3.1.15. AHB-Lite to Native Memory Bridge

This block helps the RISC-V data manager to access the True Dual-Port (TDP) memory, which is dedicated to USB operations. This block converts the AHB-Lite subordinate signal into the native memory controller interface signals.

3.2. Clocking Scheme

The CrossLinkU-NX MIPI CSI-2 to USB UVC reference design uses a PLL to generate the 72 MHz system clock, 120 MHz pixel clock for image signal processing, and 60 MHz USBPHY clock. All these clocks are generated from the 60 MHz crystal oscillator on the CrossLinkU-NX Evaluation Board.

3.2.1. Clocking Overview

An overview of the CrossLinkU-NX MIPI CSI-2 to USB UVC reference design clocking scheme is shown in [Figure 3.2](#).

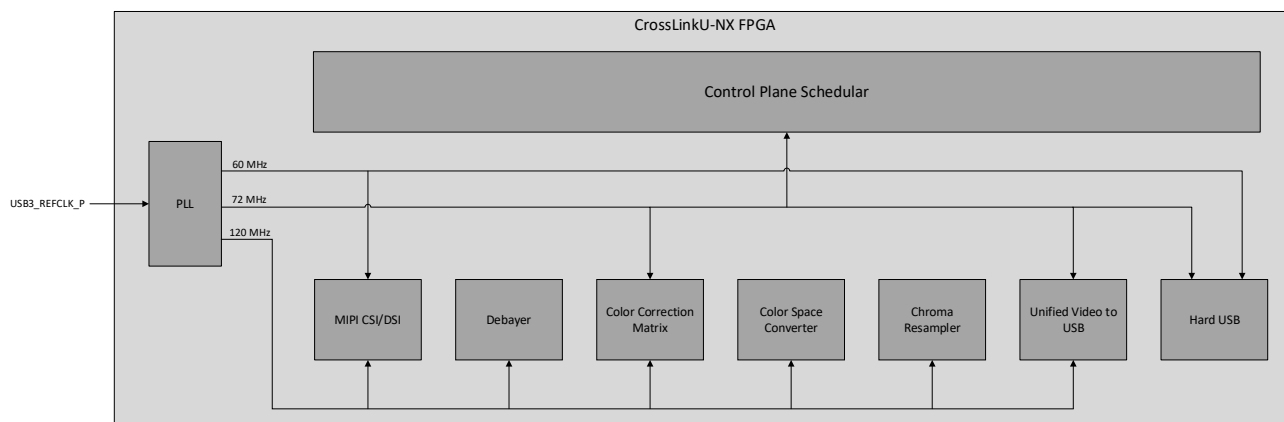


Figure 3.2. Clocking Scheme Overview

3.3. Reset Scheme

Synchronous resets are generated for each clock domain from the PLL lock signal and corresponding PLL clock outputs.

3.3.1. Reset Overview

An overview of the CrossLinkU-NX MIPI CSI-2 to USB UVC reference design reset scheme is shown in [Figure 3.3](#).

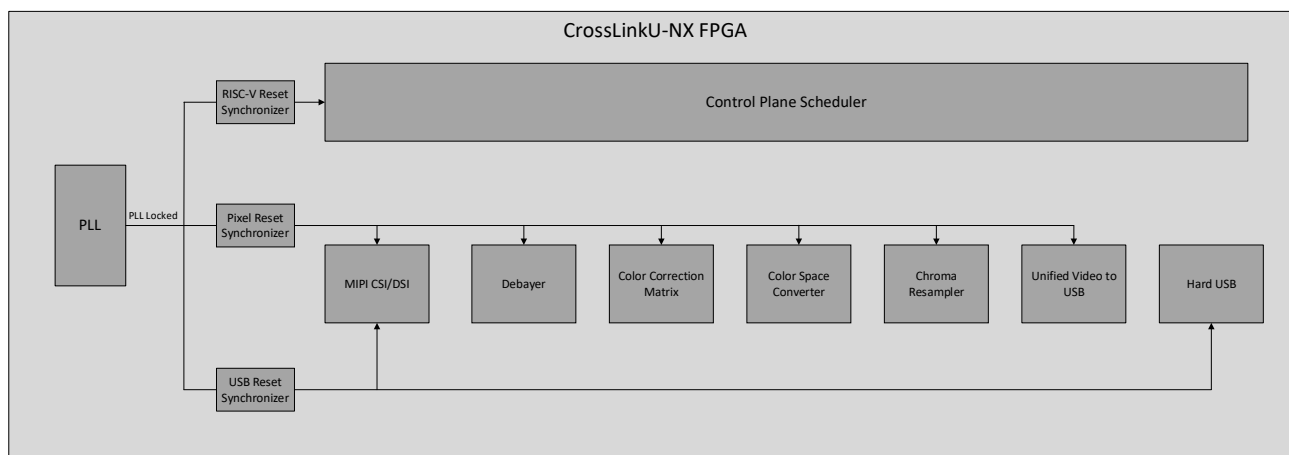


Figure 3.3. Reset Scheme Overview

4. Signal Description

Table 4.1 shows the input and output interface signals for this reference design.

Table 4.1. Primary I/O

Port Name	I/O	Width	Description
clk_60m_i	Input	1	60 MHz input clock from the on-board crystal oscillator.
dp_z	Inout	1	USB 2.0 positive differential data line
dm_z	Inout	1	USB 2.0 negative differential data line
u3_rxm_i	Input	1	USB 3.0 positive differential data line for the receiver
u3_rxp_i	Input	1	USB 3.0 negative differential data line for the receiver
vbus_z	Inout	1	Power signal
u2_reset_ext_z	Inout	1	Synchronous reset for USB 2.0
REFINCLKEXTTP_i	Input	1	USB 3.0 PHY external positive differential clock
REFINCLKEXTM_i	Input	1	USB 3.0 PHY external negative differential clock
u3_txm_o	Output	1	USB 3.0 positive differential data line for the transmitter
u3_txp_o	Output	1	USB 3.0 negative differential data line for the transmitter
cam_scl_z	Inout	1	I2C controller serial clock line (SCL) for the camera sensor
cam_sda_z	Inout	1	I2C controller serial data line (SDA) for the camera sensor
cam_en_o	Output	1	Camera sensor enable
rx_clk_p_i	Inout	1	Camera positive differential clock line
rx_clk_n_i	Inout	1	Camera negative differential clock line
rx_d0_p_i	Inout	1	Camera positive differential data0 line
rx_d0_n_i	Inout	1	Camera negative differential data0 line
rx_d1_p_i	Inout	1	Camera positive differential data1 line
rx_d1_n_i	Inout	1	Camera negative differential data1 line
uart_rxd_i	Input	1	UART input for RISC-V microcontroller debugging
uart_txd_o	Output	1	UART output for RISC-V microcontroller debugging

5. Running the Reference Design

This section describes the steps to run the reference design in the Lattice Propel SDK, Lattice Propel Builder, and Lattice Radiant software. If you want to evaluate the reference design using the precompiled bitstream, you can skip the build steps in this section and proceed directly to the [Precompiled Bitstream](#) section.

Note: The screenshots provided are for reference only. Details may vary depending on the version of the IP, reference design, or software being used. If there have been no significant changes to the GUI, a screenshot may reflect an earlier version of the IP and reference design.

5.1. Running Propel SDK Project

This reference design uses the hardened USB block inside the LIFCL-33U device. The block is configured and USB enumeration is performed by the RISC-V firmware. This section describes how to build and run a C project in the Lattice Propel SDK software.

To run the reference design in the Lattice Propel SDK software, follow these steps:

1. Launch the Lattice Propel SDK software.
2. Click the **Browse** button to set the workspace folder to
<Project Directory>/rd02306_mipi_csi_to_usb_uvc/fpga_lifcl/propelsdk.
3. Then, click the **Launch** button, as shown in [Figure 5.1](#).

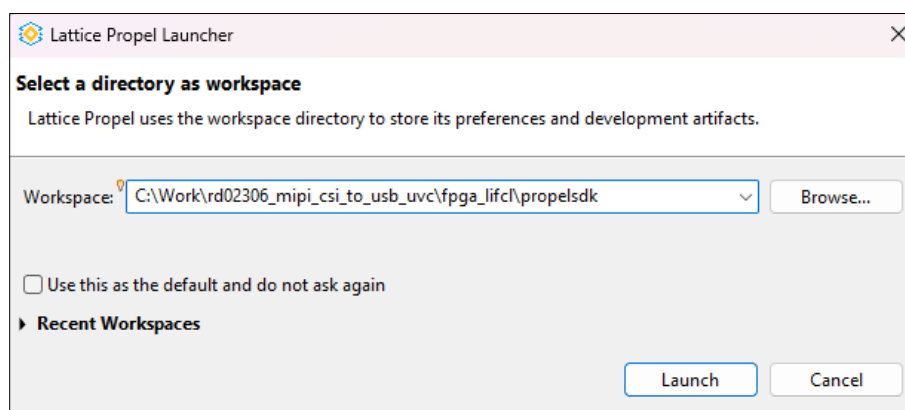


Figure 5.1. Launch Lattice Propel SDK

4. In the GUI, click **Import projects...**, and then select **Existing Projects into Workspace** under **General** ([Figure 5.2](#)).

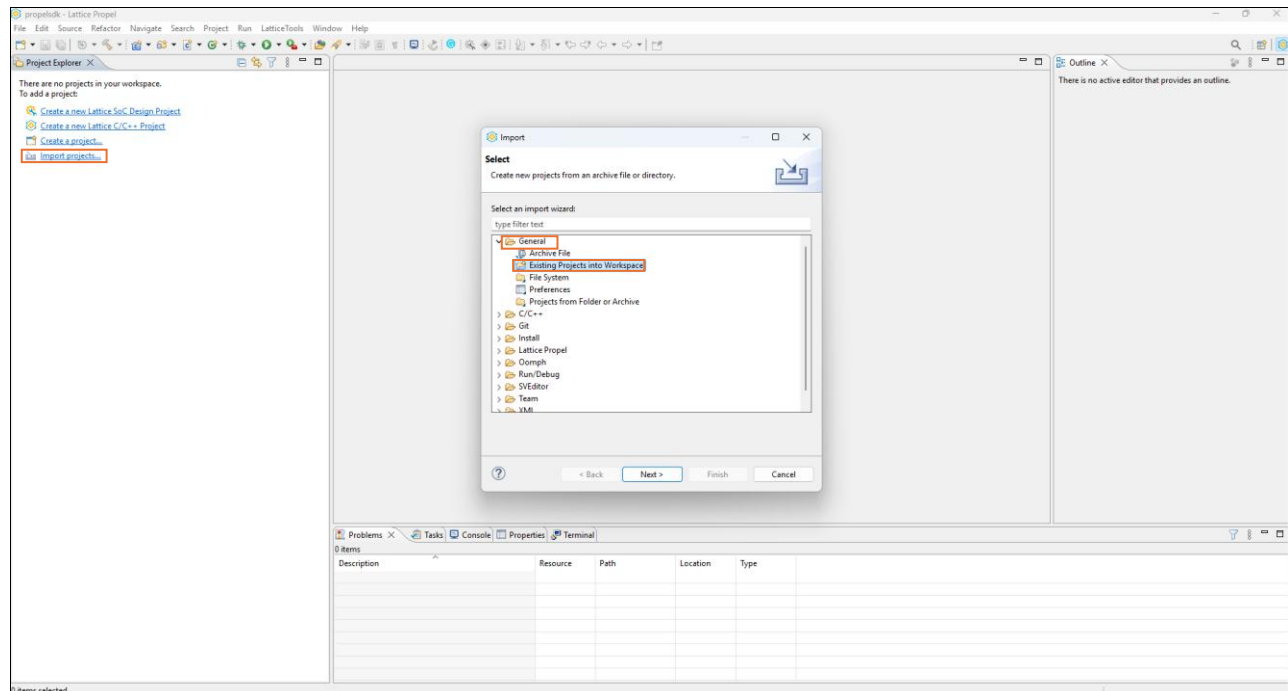


Figure 5.2. Import a Project in Lattice Propel SDK

- Click the **Browse** button to set the root directory (Figure 5.3):
<Project Directory>/rd02306_mipi_csi_to_usb_uvc/fpga_lifcl/propelsdk/riscv_mc.
Then, click **Finish**.

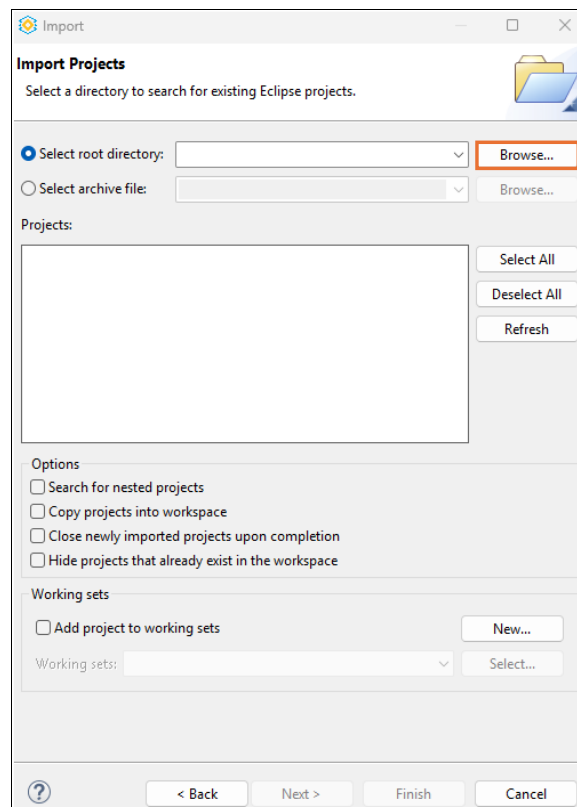


Figure 5.3. Browse Projects Root Directory

6. The Propel SDK project opens and you can see the riscv_mc project (Figure 5.4).

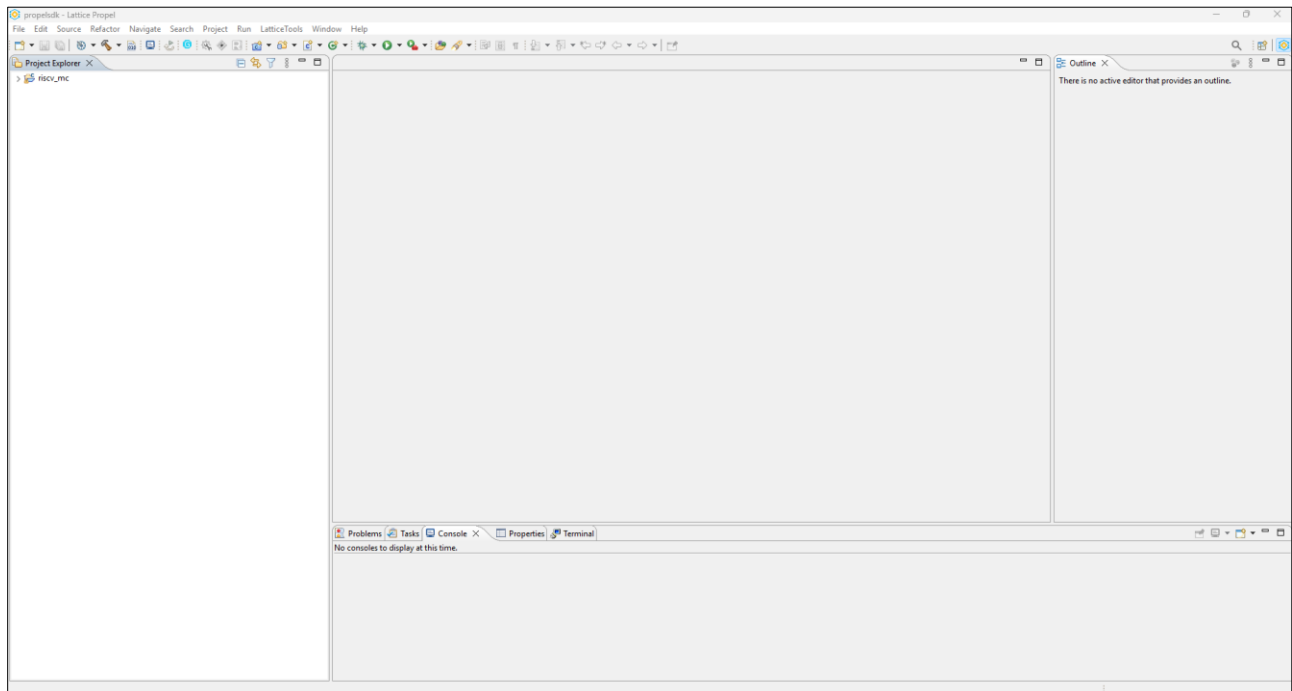


Figure 5.4. Propel SDK Project Opened in Lattice Propel SDK

7. Navigate to **riscv_mc > src** to expand the project file list, as shown in Figure 5.5.

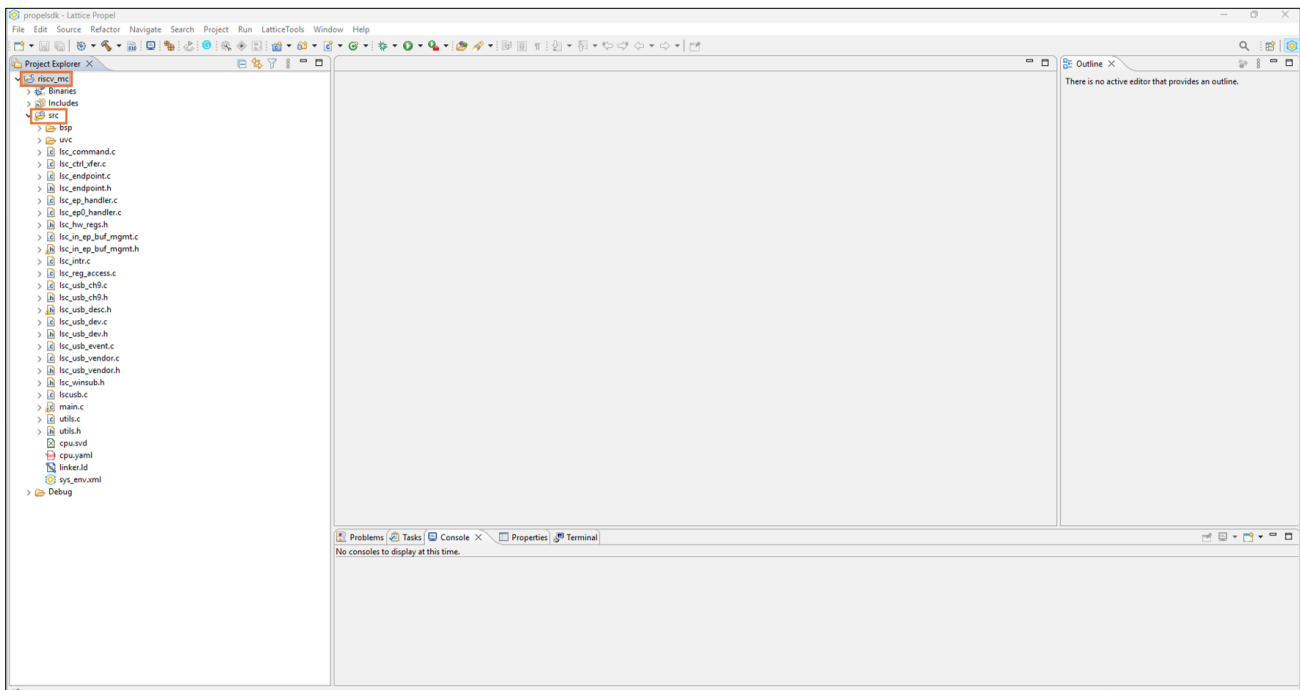


Figure 5.5. Navigate the Propel SDK Project

8. Select **Project > Build Configurations > Set Active > 1 Debug** (Figure 5.6) to compile the Propel SDK project to generate the memory image for the RISC-V firmware.

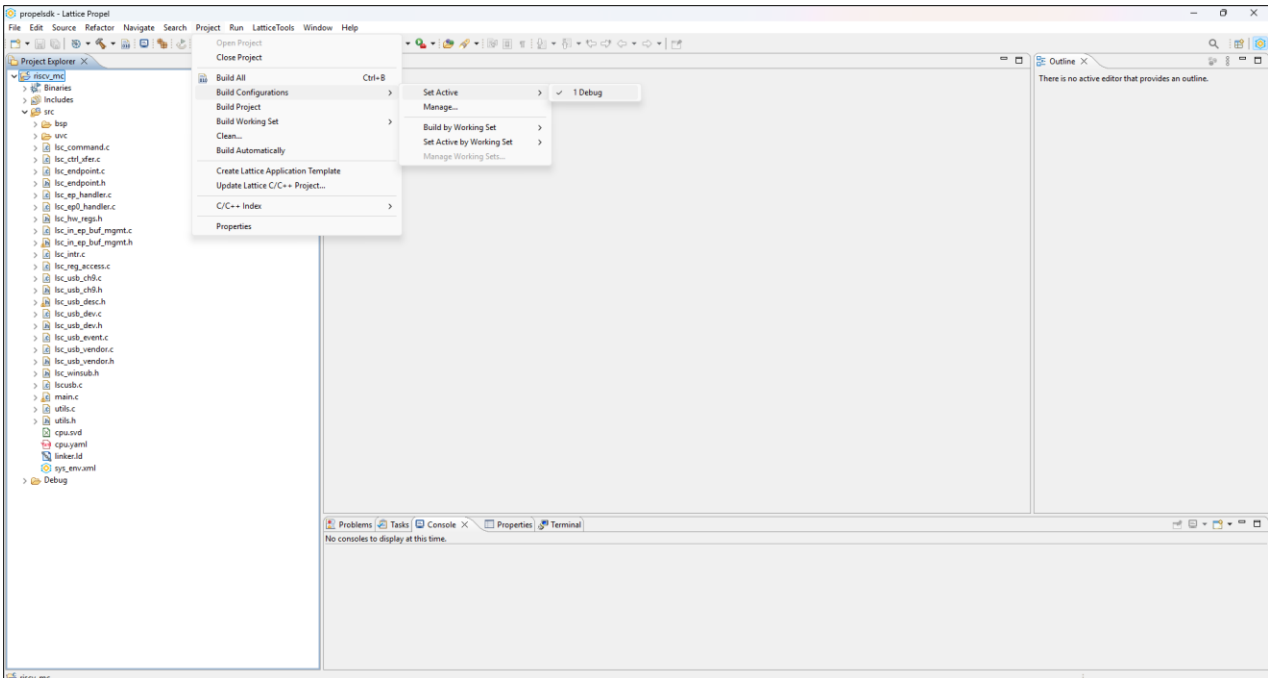


Figure 5.6. Set Propel Build Configuration Mode

- To launch the project building process, right-click on the process name **riscv_mc**, select **Clean Project** from the menu to clean up the output folders. Then, select **Build Project** from the menu, as shown in Figure 5.7. The output riscv_mc.mem file is located at:
<Project Directory>/rd02306_mipi_csi_to_usb_uvc/fpga_lifc1/propelsdk/riscv_mc/Debug/riscv_mc.mem.

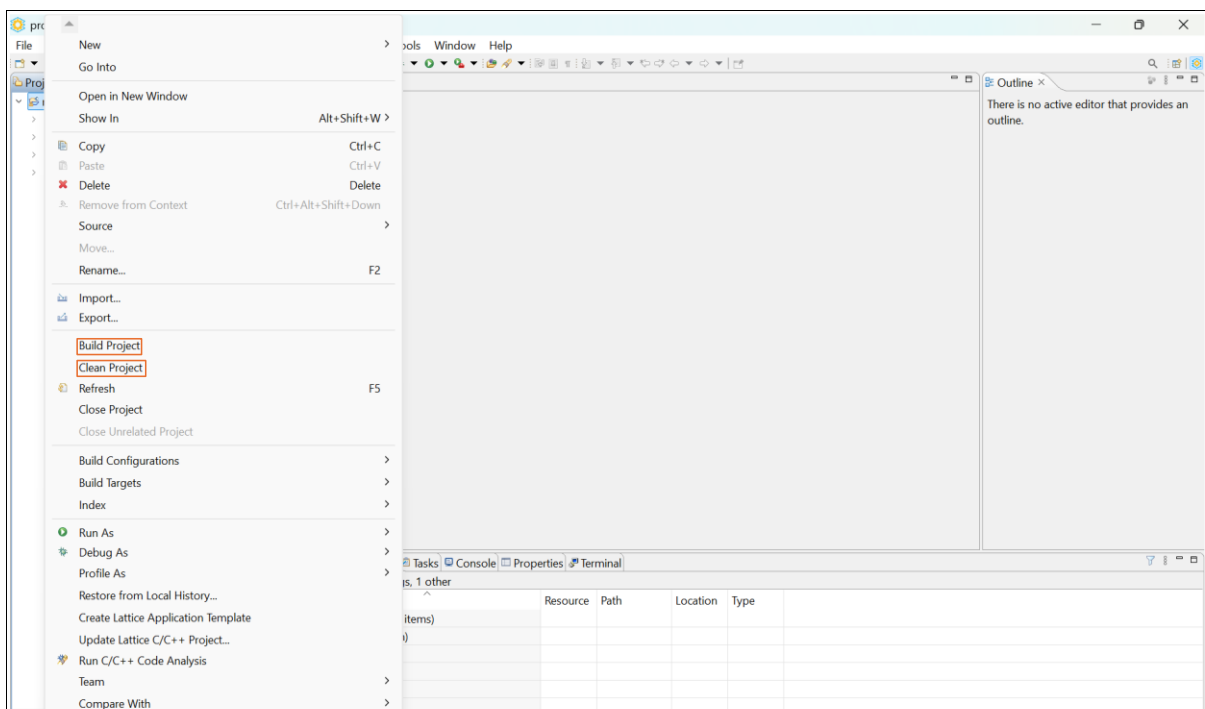


Figure 5.7. Launch the Project Building Process

5.2. Running Propel Builder and Radiant Project

This section describes the steps to run the reference design in the Lattice Propel Builder and Lattice Radiant software.

To run the reference design in the Lattice Propel Builder and Lattice Radiant software, follow these steps:

1. Launch the Lattice Propel Builder software.
2. Go to **File > Open Design**, browse and select the existing project file: *mipi_csi_to_usb_uvc.sbx*.

Note: The expected INFO and WARNING messages appear after opening the project file. Refer to the [Debugging](#) section for a detailed explanation of these messages.

3. Before performing any IP reconfiguration, ensure that the IP has been downloaded and installed under **IP on Local**, as shown in [Figure 5.8](#). In this reference design, there are a few IPKs that need to be installed into **IP on Local**. Follow the steps below to perform the IPK installation.
 - a. Right-click **IP on Local** and select **Install IP from file...**, as shown in [Figure 5.9](#).
 - b. In the pop-up file browser, select the following IPK path:
<Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/
latticesemi.com_ahbl_to_axi4lite_bridge_1.0.0.0.ipk.
 - c. Click **OK**.
 - d. In the **IP License Agreement** window, click **Accept** to finish the IP installation, as shown in [Figure 5.10](#).
 - e. Repeat Steps a to d above to install the rest of the IPKs listed below:
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/
latticesemi.com_ahbl_to_lmmbi_bridge_1.1.0.0.ipk.
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/
latticesemi.com_ahbl_to_mem_bridge_1.0.0.0.ipk.
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_uvsi_
to_usb_1.0.0.0.ipk
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/
latticesemi_custom_usb23_bus_intf_bridge_1.0.0.0.ipk
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_ccm_1
.3.1.03.ipk
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_color_
space_converter_2.4.1.03.ipk
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_debay
er_1.4.0.03.ipk
 - <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_chrom
a_resampler_1.0.0.0.ipk
 - f. After completing the IPK installation, the IP mentioned above should be available under **IP on Local**, as shown in [Figure 5.11](#).

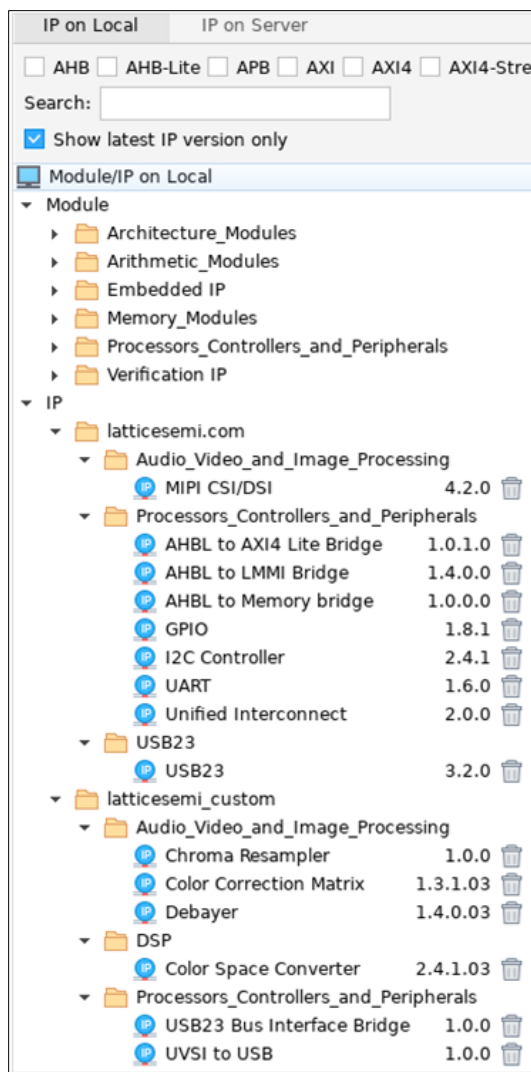


Figure 5.8. IP on Local Tab in IP Catalog

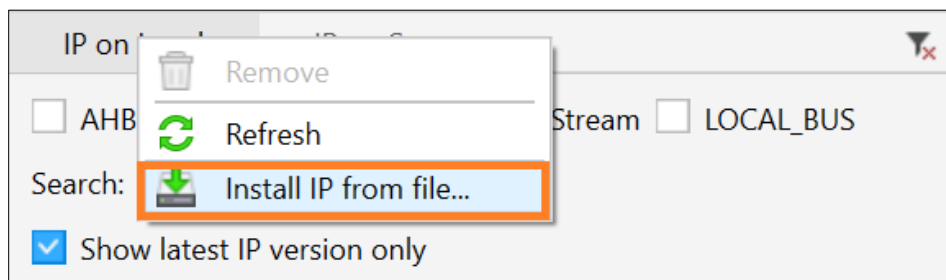


Figure 5.9. Install IP from File

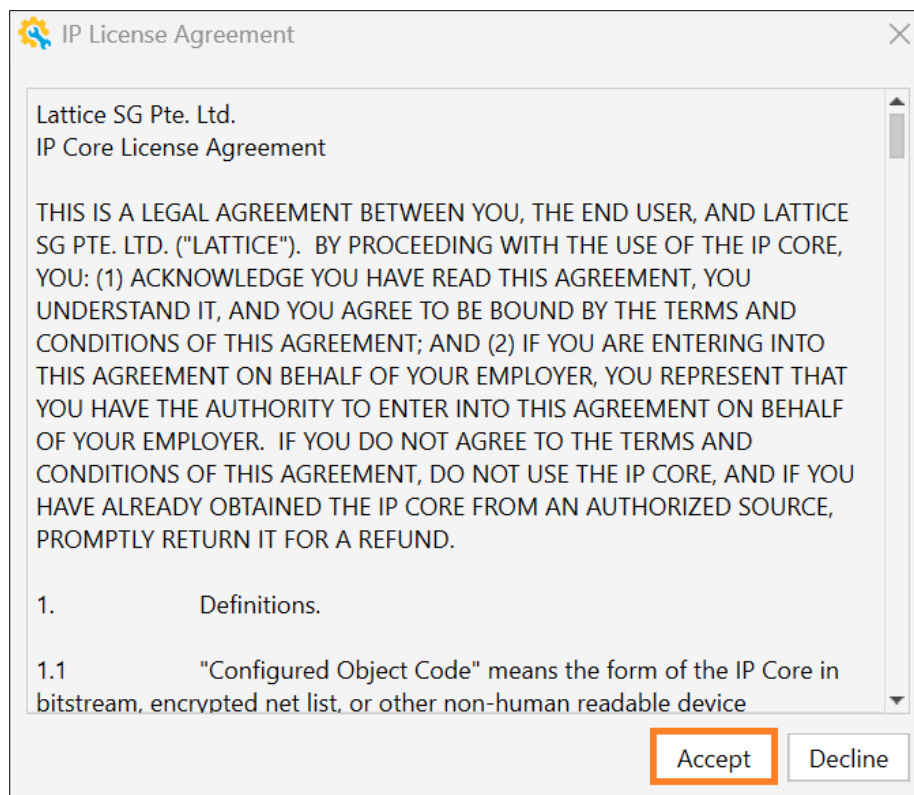


Figure 5.10. IP License Agreement

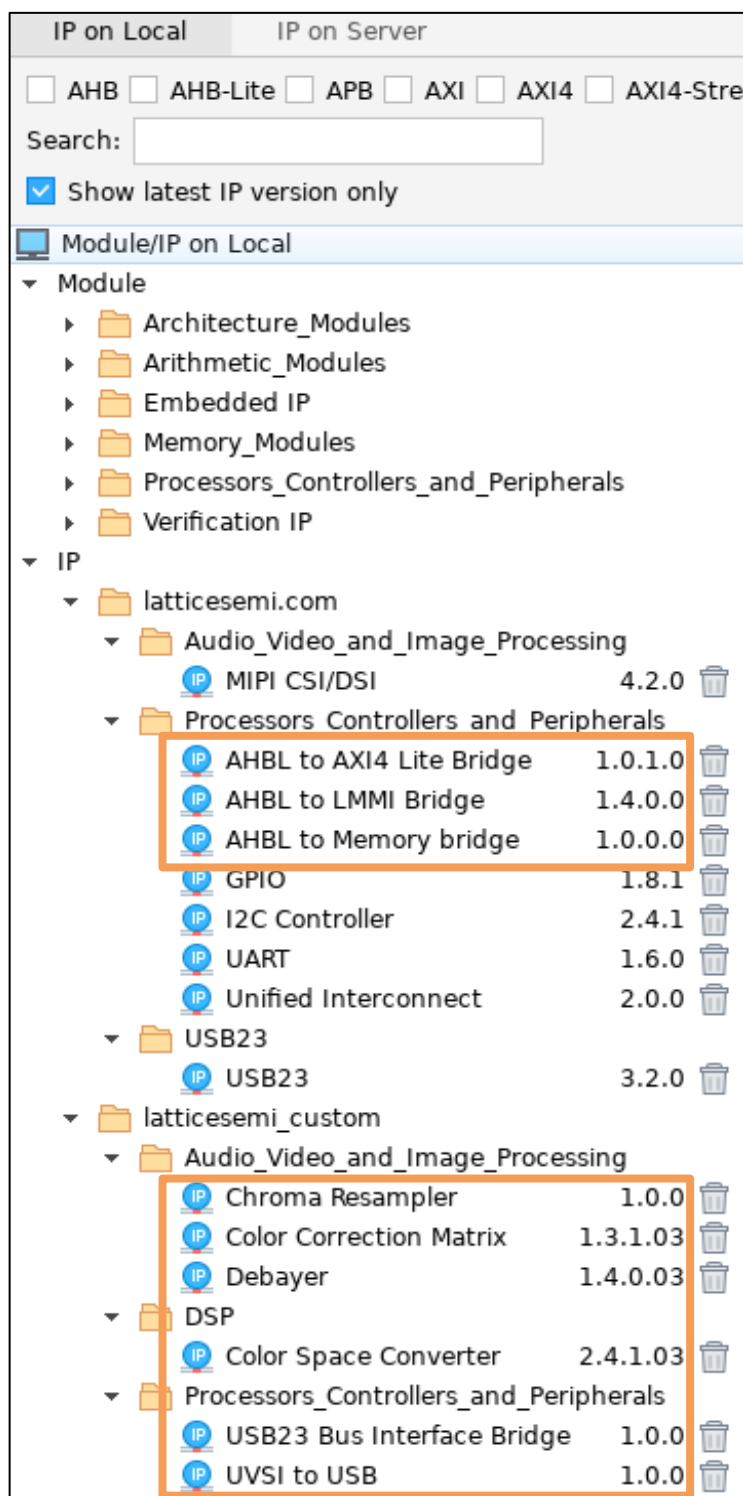


Figure 5.11. IPs Installed from Files

- In the Propel Builder design schematic, double-click `system0_inst` to open the `system_memory` Module/IP Block Wizard and ensure the Initialization File is mapped to `../../propelsdk/riscv_mc/Debug/riscv_mc.mem` or the latest revision of the .mem file. Then, click the **Generate** button, as shown in Figure 5.12.

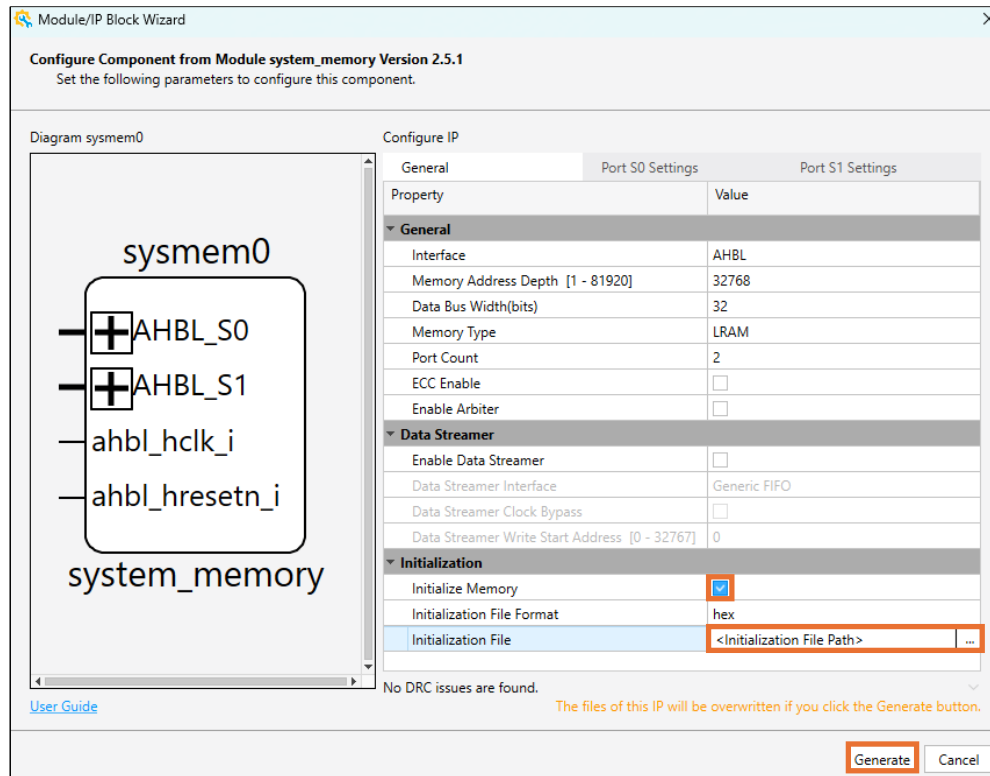


Figure 5.12. Mapping the System Memory Initialization Files

- Click the **Address** tab to review the base address assignment for the system and all peripherals within the Propel Builder Project, as shown in Figure 5.13.

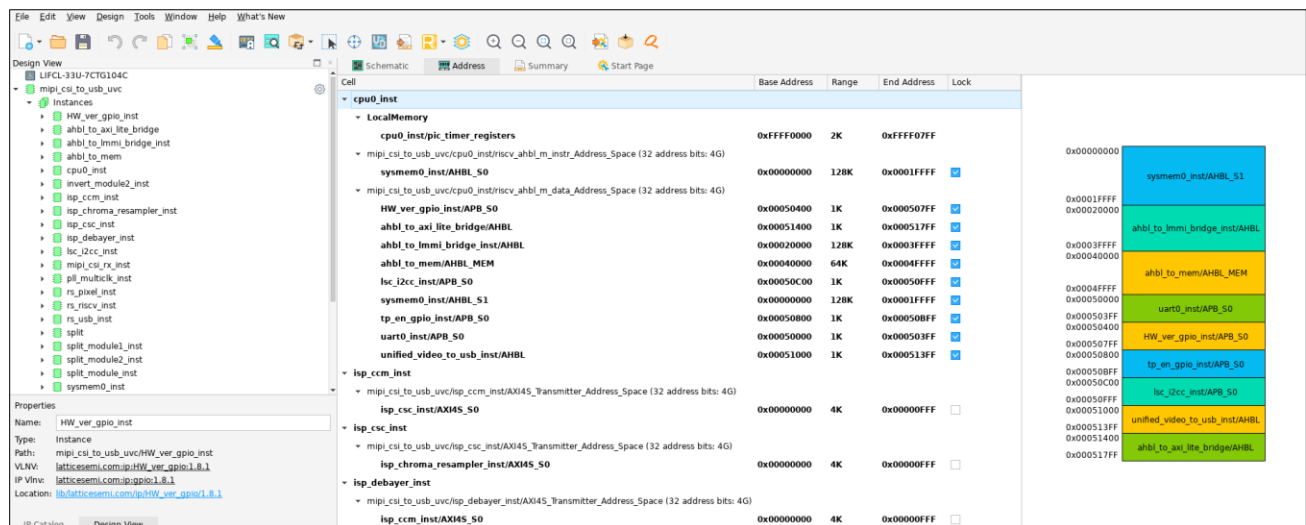


Figure 5.13. Base Address Assignment

Note: To reconfigure the IP, refer to the [Customizing the Reference Design](#) section for the available IP configuration options.

- To validate the design, select **Design > Validate Design** or click the **Validate Design** icon (as shown in Figure 5.14).

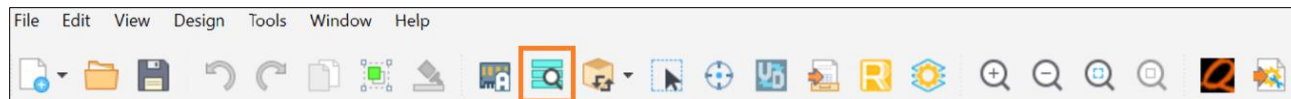


Figure 5.14. Validate the Design in Lattice Propel Builder

7. To generate the design, select **Design > Generate** or click the **Generate** icon (as shown in [Figure 5.15](#)).

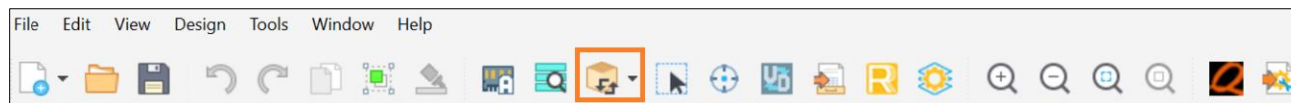


Figure 5.15. Generate the Design in Lattice Propel Builder

Note: Ensure the TCL console shows no errors, and only the expected INFO and WARNING messages appear. Refer to the Debugging section for a detailed explanation of these messages.

8. To synthesize the RTL files and generate the bitstream, click the Lattice Radiant icon in Lattice Propel Builder, as shown in [Figure 5.16](#).

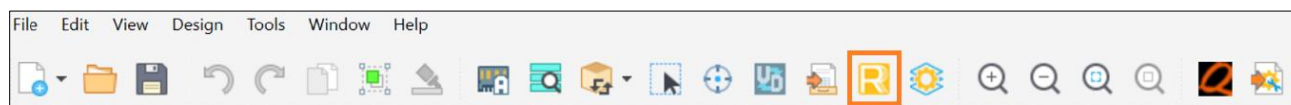


Figure 5.16. Lattice Radiant Icon in Lattice Propel Builder

9. The Lattice Radiant software window opens and shows this project ([Figure 5.17](#)).

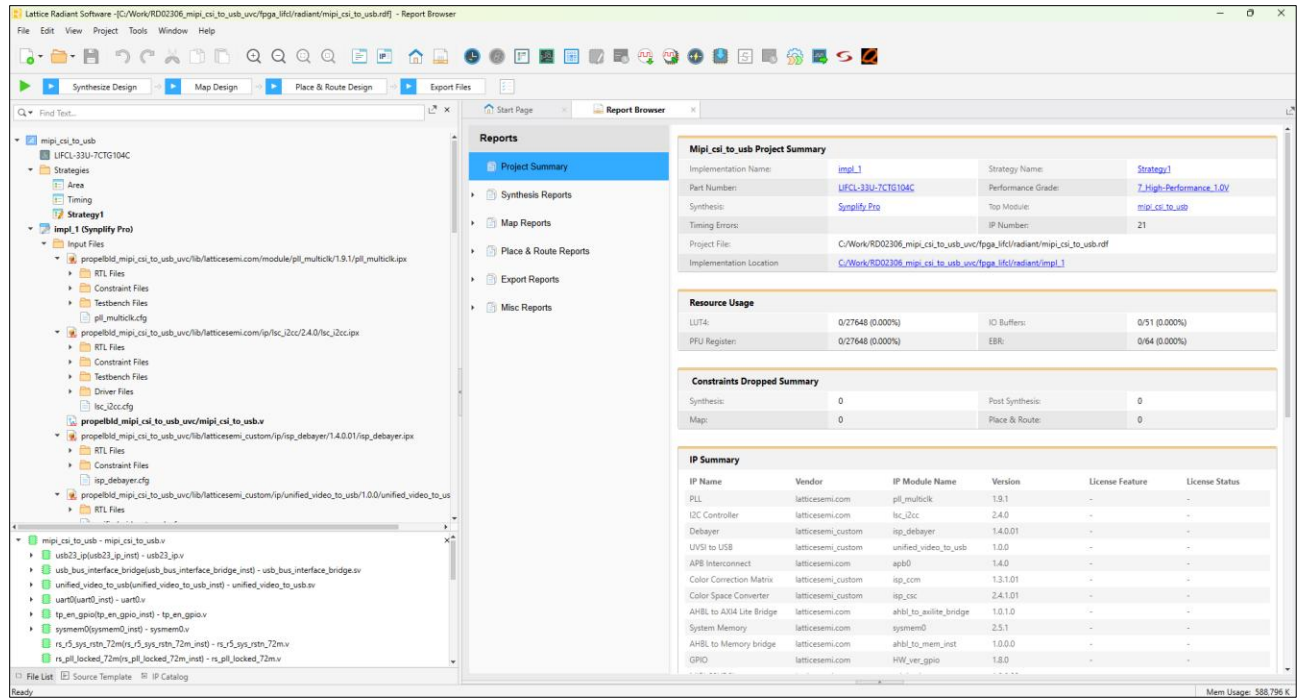


Figure 5.17. Lattice Radiant Software Window

10. Set the synthesis strategy (Figure 5.18) and the Place and Route design strategy (Figure 5.19).

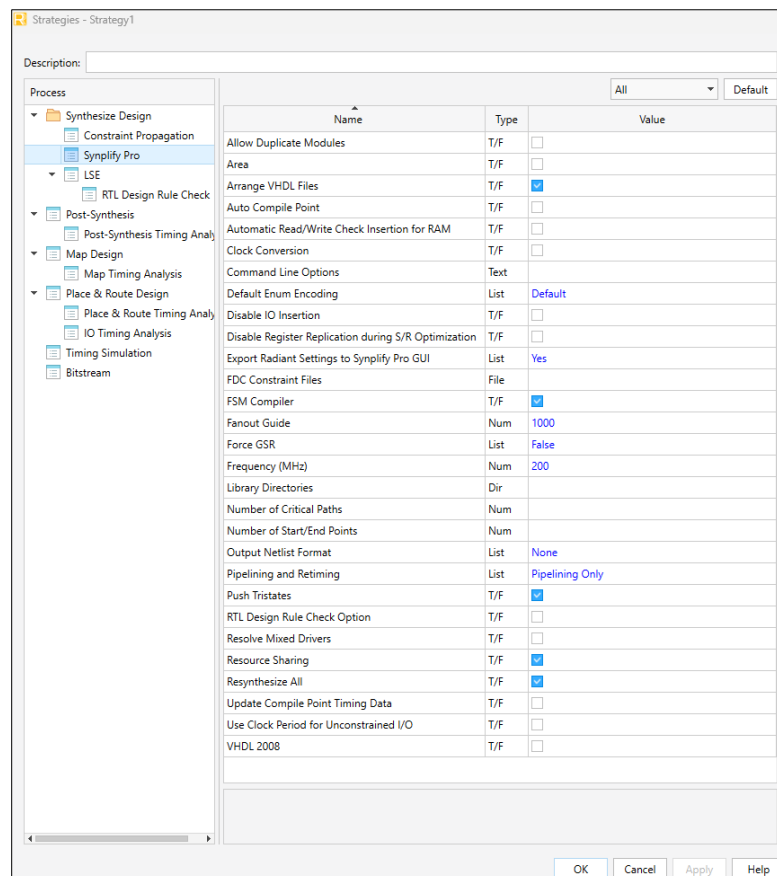


Figure 5.18. Synthesis Strategy Settings

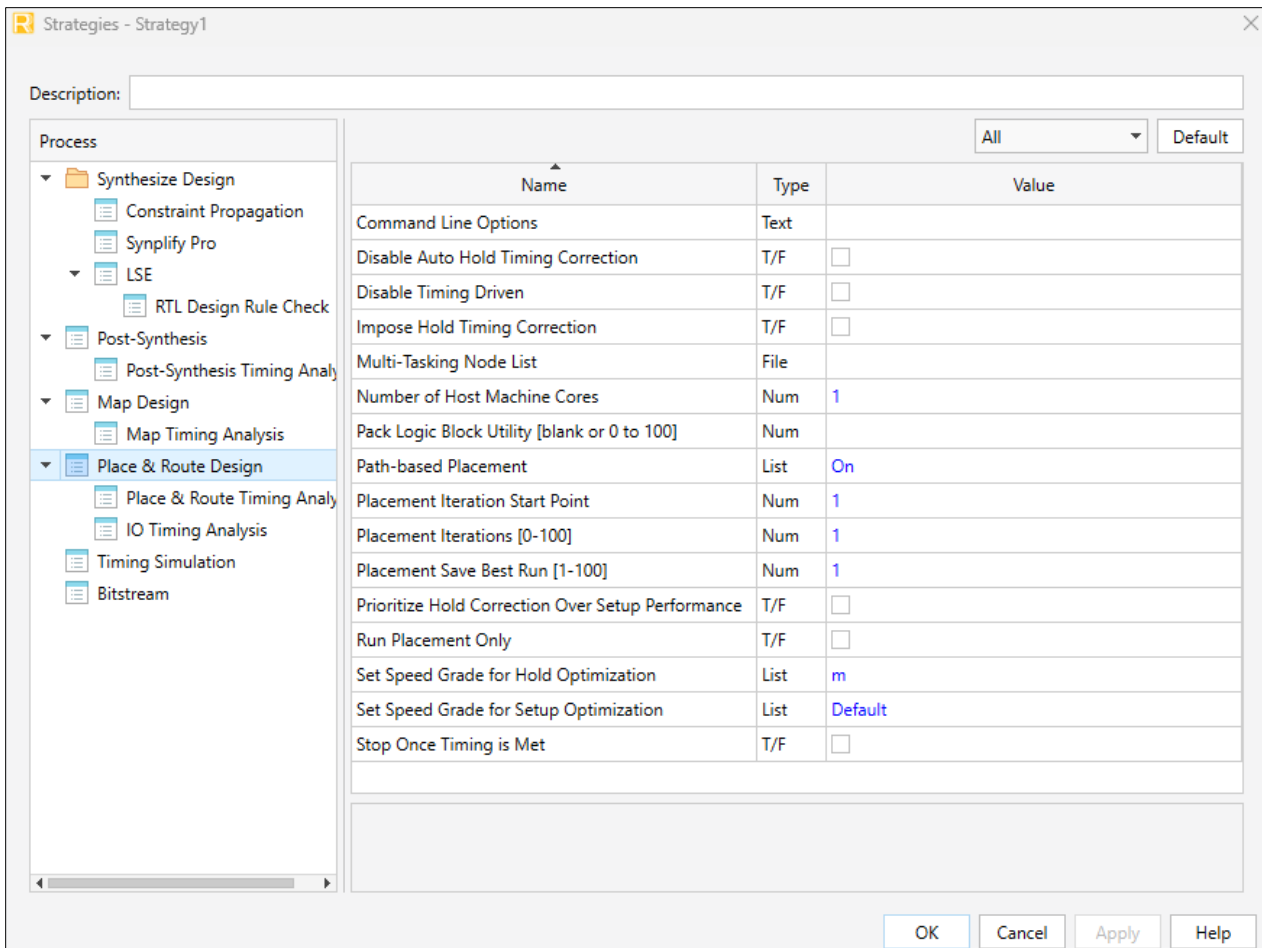


Figure 5.19. Place and Route Design Strategy Settings

- Click the **Run All** icon to generate the bitstream (Figure 5.20).

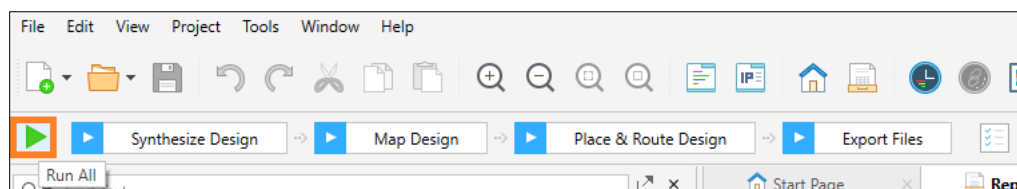


Figure 5.20. Generating Bitstream

- The compilation flow takes some time to complete. As shown in Figure 5.21, when the compilation is completed successfully, the `mipi_csi_to_usb_uvc_impl_1.bit` file is generated or updated in the project path:
`<Project Directory>/rd02306_mipi_csi_to_usb_uvc/fpga_lifcl/radiant/impl_1.`



Figure 5.21. Compilation Completed

Note: If timing violations occur or Place and Route fails during compilation or fails to generate a bitstream, refer to the [Debugging](#) section for detailed debugging instructions.

6. Implementing the Reference Design on Board

This section describes how to run this reference design on the CrossLinkU-NX Evaluation Board.

6.1. Programming the FPGA

6.1.1. Recompiled Bitstream

To program the recompiled bitstream, follow these steps:

1. Connect the Micro USB port (J2) of the CrossLinkU-NX Evaluation Board to your PC by using a Micro USB cable.
2. Click the **Programmer** icon, as shown in Figure 6.1.

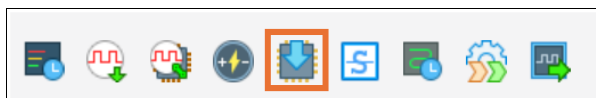


Figure 6.1. Programmer Icon

3. The main interface opens, as shown in Figure 6.2.

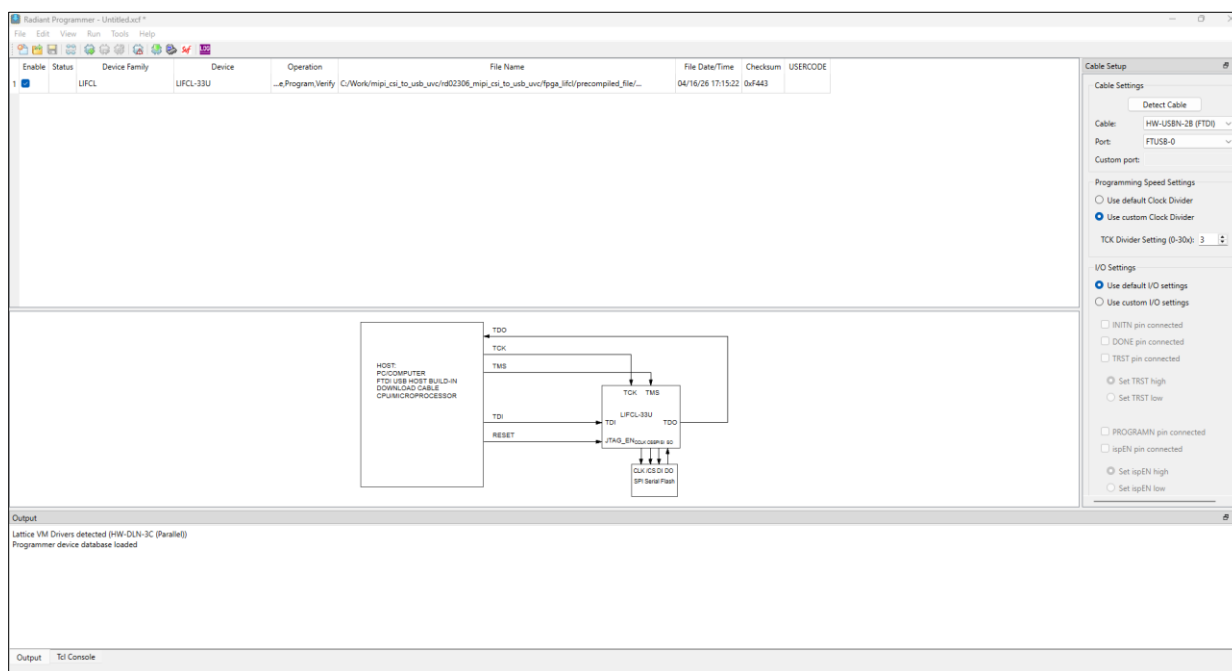


Figure 6.2. Lattice Radiant Programmer Window

To select the programming settings, follow these steps:

- a. Double-click **Operation** to open the **Device Properties** window.
- b. The generated bitstream file is automatically selected: <Project Directory>/rd02306_mipi_csi_to_usb_uvc/fpga_lifcl/radiant/impl_1/mipi_csi_to_usb_uvc_impl_1.bit.
- c. Select the other settings shown in Figure 6.3 and click **OK**.

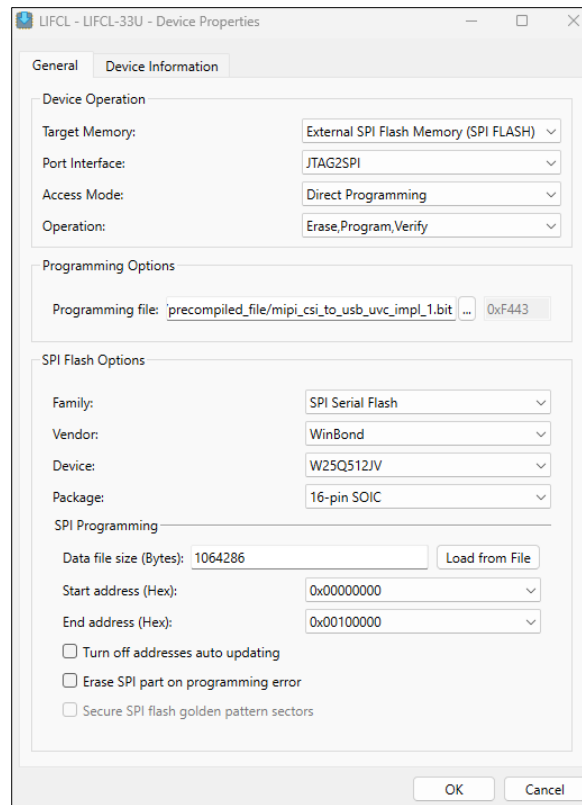


Figure 6.3. Device Properties Window for CrossLinkU-NX SPI Flash Programming

- Click the **Program Device** button from the menu bar to start programming (Figure 6.4).

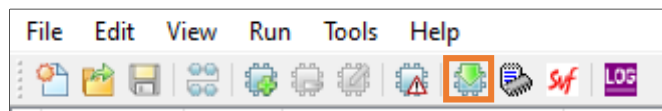


Figure 6.4. Programming Menu Bar

When the FPGA programming is successful, the output console displays the **Operation: successful** message.

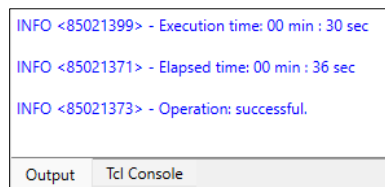


Figure 6.5. Programmer Output Window

Note: If the programming operation is unsuccessful, refer to the [Debugging](#) section.

- After programming, unplug the programming cable to ensure that the board is not powered. Then, follow the steps in the [Hardware Setup](#) section to set up the hardware.

6.1.2. Precompiled Bitstream

To program the precompiled bitstream, follow these steps:

1. Create a new project using the Lattice Radiant Programmer software. In the **Getting Started** dialog box, input the **Project Name** and **Location**, as shown in [Figure 6.6](#).
2. Select **Create a new project from a scan**. The values are indicated in the **Cable**, **Port**, and **TCK Divider Setting (0–30x)** fields.
3. Click **OK**.

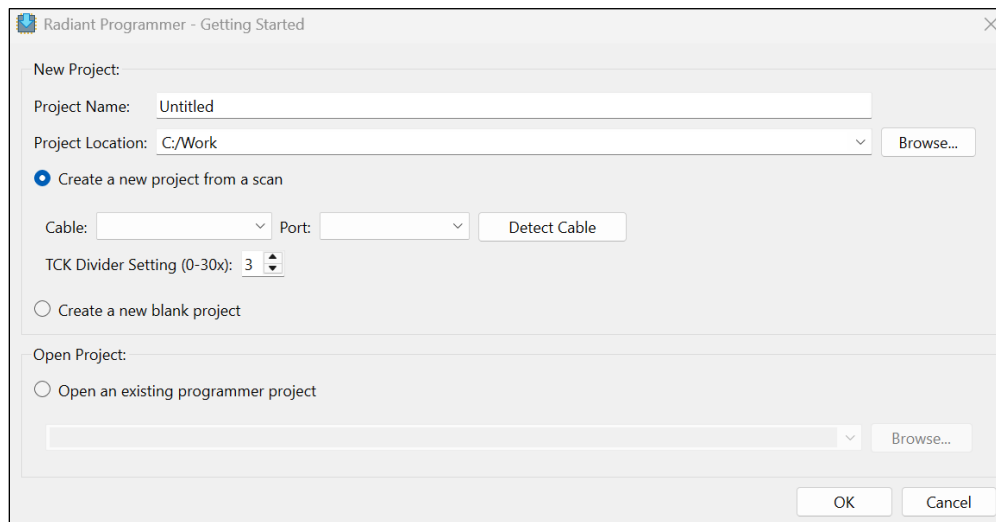


Figure 6.6. Creating a New Project from a Scan

Note: To open an existing programmer project, select **Open an existing programmer project** and browse to the project file (.xcf).

4. The Radiant Programmer main interface opens, as shown in [Figure 6.7](#).

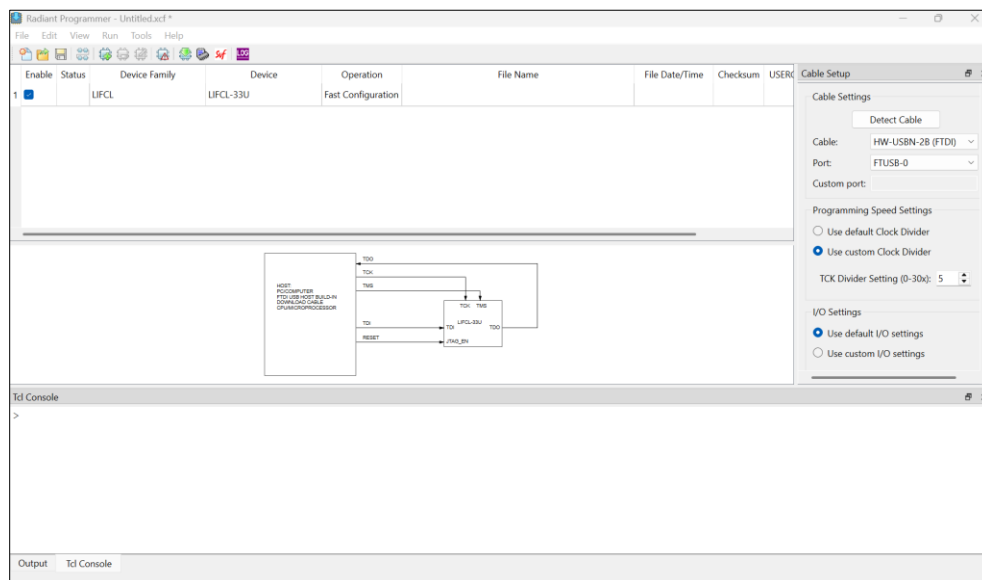


Figure 6.7. Lattice Radiant Programmer Window

5. If the Programmer settings do not match the settings for CrossLinkU-NX devices shown in [Figure 6.8](#), select these settings manually from the drop-down menu.

Enable	Status	Device Family	Device	Operation	File Name	File Date/Time	Checksum	USERCODE
☒		LIFCL	LIFCL-33U	Fast Configuration				

Figure 6.8. CrossLinkU-NX Device Settings

6. To select the programming settings, follow these steps:
 - a. Double-click Operation to open the Device Properties dialog box.
 - b. Browse and select the precompiled bitstream file, mipi_csi_to_usb_uvc_impl_1.bit.
 - c. Select the other settings, as shown in Figure 6.9.

Note: If the flash device is a GigaDevice flash, then the SPI Flash options are GD25LQ128E and 8-pin SOP. If the flash device is a Winbond flash, then the SPI Flash options are W25Q512JV and 8-pad WSON.

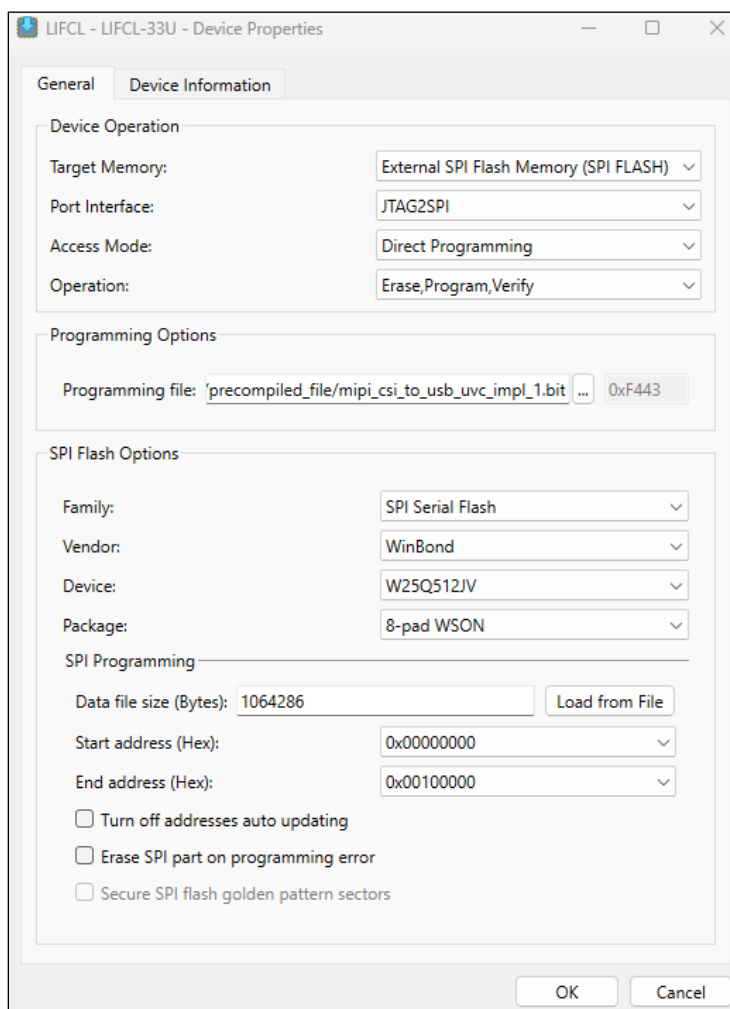


Figure 6.9. Device Properties Window for the CrossLinkU-NX SPI Flash Programming

7. Click **Programming** from the menu bar to start programming (Figure 6.10).

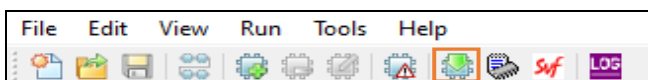


Figure 6.10. Programmer Menu Bar

8. When the FPGA programming is successful, the output console displays the **Operation: successful** message (Figure 6.11).

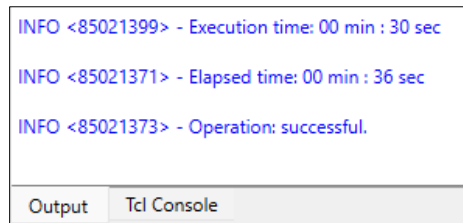


Figure 6.11. Programming Output Window

Note: If the programming operation is unsuccessful, refer to the [Debugging](#) section.

6.2. Hardware Setup

Connect the Raspberry Pi Camera Module V2 to the CN1 connector on the CrossLinkU-NX Evaluation Board, which is the camera connector. Connect the USB Type-C connector CN2 on the pre-programmed CrossLinkU-NX Evaluation Board to the PC using the USB Type-C to USB Type-C cable or the USB Type-C to USB Type-A 9-pin cable, as shown in Figure 6.12. The device is enumerated as Lattice USB23, which can be found in Windows Device Manager, as shown in Figure 6.13.

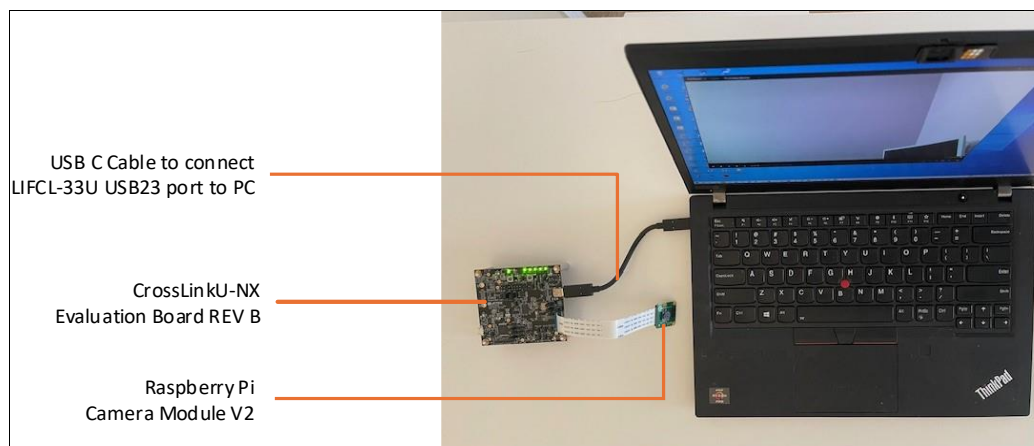


Figure 6.12. CrossLinkU-NX Evaluation Board Setup for Demonstration

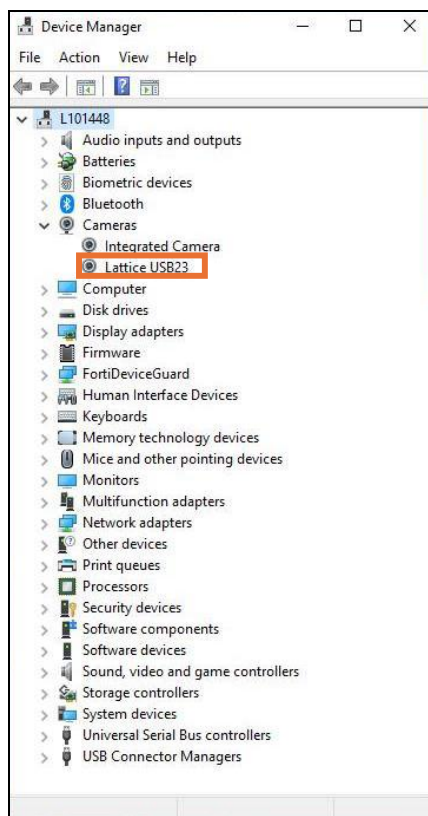


Figure 6.13. USB Enumeration

6.3. Demonstration

To demonstrate this reference design, follow these steps:

1. Prior to running this demonstration, ensure PotPlayer is installed. This reference design uses version 1.7.
2. Open the PotPlayer application.
3. To open a video, either press the shortcut Ctrl+J or click **PotPlayer > Open > Open Webcam/Other Device**, as shown in Figure 6.14.

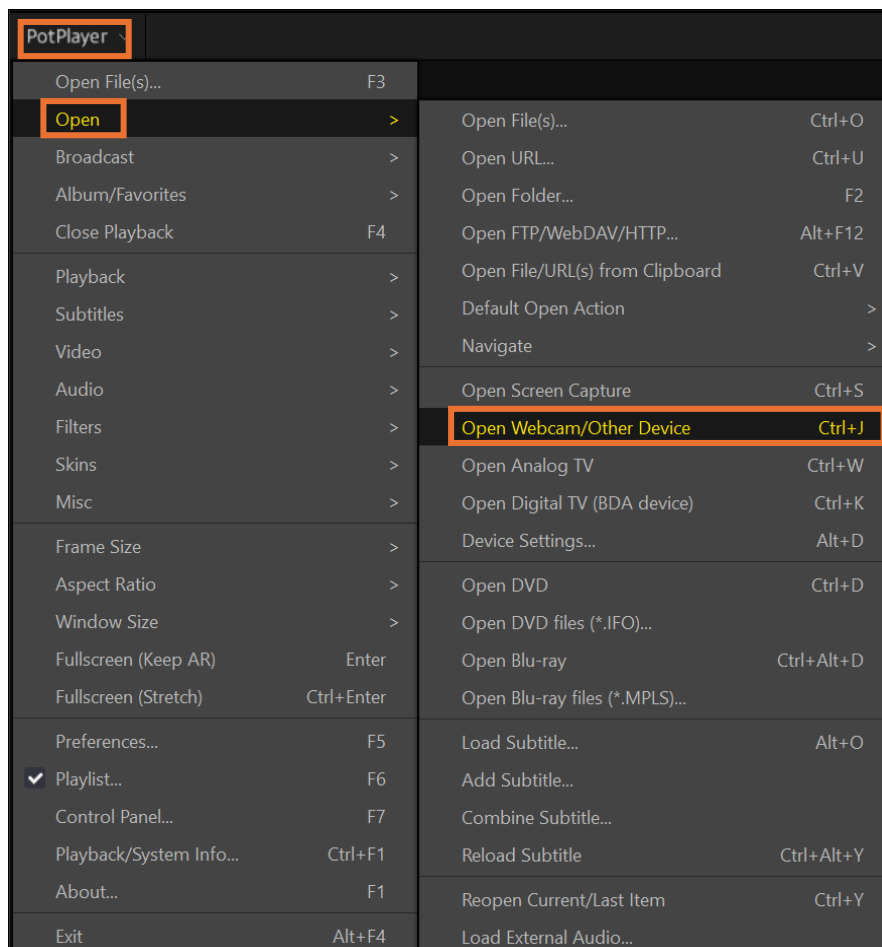


Figure 6.14. Open Webcam or Other Devices in PotPlayer

4. The video of the camera sensor can be seen on the PotPlayer screen, as shown in Figure 6.15.



Figure 6.15. Video Output on PotPlayer

5. To change the Video Renderer option in PotPlayer, right-click on the video and select **Video > Video Renderer > Built-in Direct3D 11 Video Renderer**, as shown in Figure 6.16.

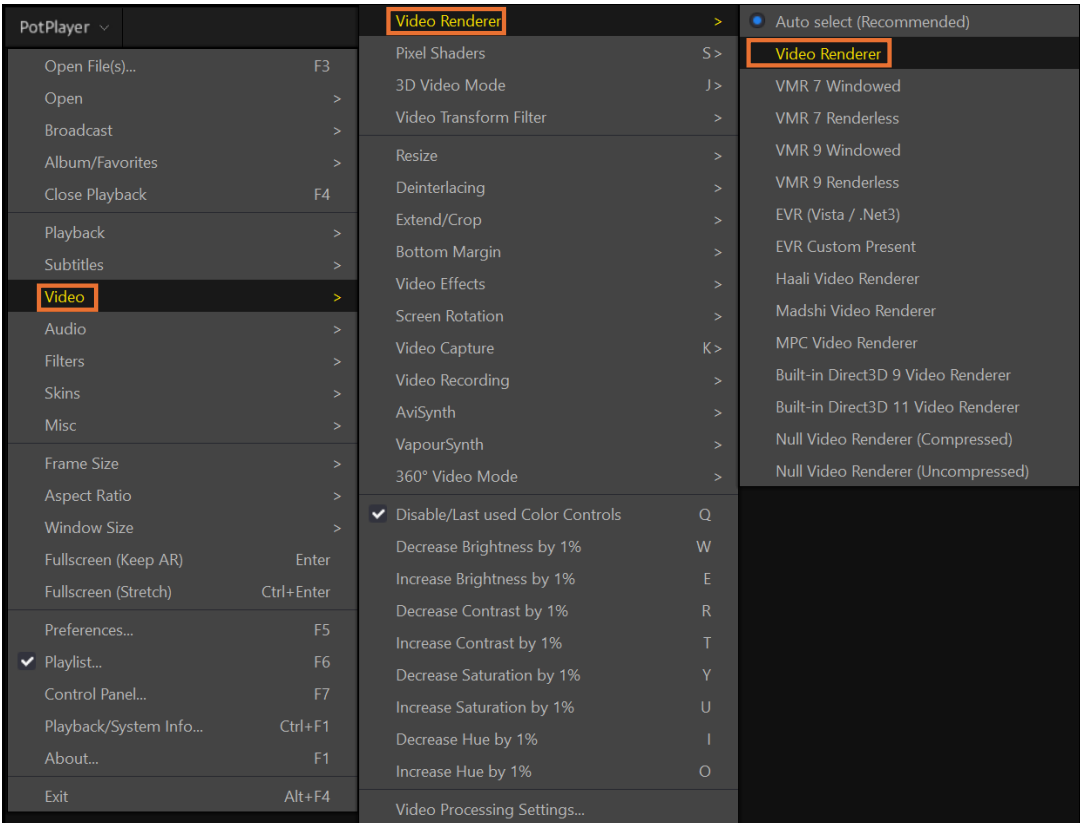


Figure 6.16. Video Renderer Option

6. To view the frames per second (FPS), either press the shortcut Ctrl+F1 or click **PotPlayer > Playback/System Info...**, as shown in Figure 6.17. An example of the video information is shown in Figure 6.18.

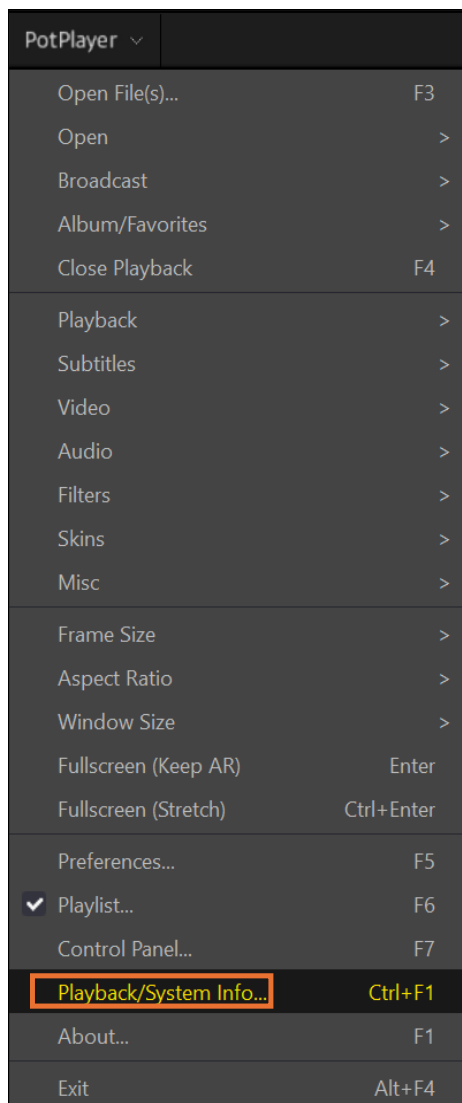


Figure 6.17. Playback/System Info...

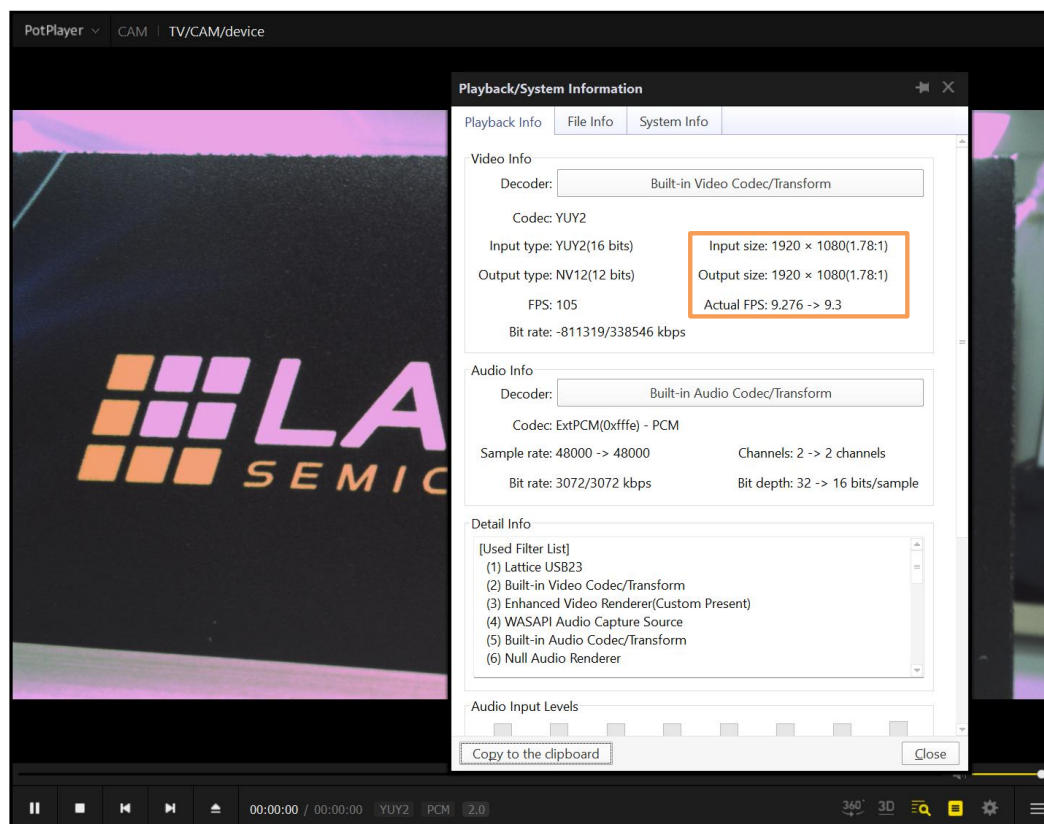


Figure 6.18. Example of Video Information

Note: If no display appears after programming the precompiled or recompiled bitstream or the display turns black when switching resolution during active display, refer to the [Debugging](#) section for detailed debugging instructions.

7. Customizing the Reference Design

This section describes the steps required to configure each IP and software for various demonstration setups, starting from the default design configuration. It also provides a way to enable or disable the YUV422 test pattern generator.

While the default IP configuration can support all resolutions and frame rates, the IP can be reconfigured to demonstrate a lower pixel-per-clock setup, resulting in reduced resource utilization. The IMX219 sensor is configured to operate at a 592 Mbps link rate with a 2-lane configuration. Refer to [Table 7.1](#) for the IP configuration setting according to the desired setup.

Table 7.1. IP Configuration for Supported Resolutions

Resolution/FPS	USB Mode	D-PHY Link Rate (Mbps)	D-PHY Lane Count ^{1,2}	Pixels Per Clock Required ^{3,4} (Clocking at 120 MHz)
3280x2160/15.61	3.0 (SuperSpeed)	592	2	1 or 2
1920x1080/30.66	3.0 (SuperSpeed)	592	2	1 or 2
1280x720/45.19	3.0 (SuperSpeed)	592	2	1 or 2
640x480/66.05	3.0 (SuperSpeed)	592	2	1 or 2
1920x1080/9.30	2.0 (High-Speed)	592	2	1 or 2
1280x720/20.56	2.0 (High-Speed)	592	2	1 or 2
640x480/60.11	2.0 (High-Speed)	592	2	1 or 2

Notes:

1. IMX219 does not support a 4-lane configuration.
2. The driver does not support a 1-lane configuration.
3. Pixels Per Clock of 4 is not supported due to high device resource requirements.
4. The required clock frequency of the pixel clock is the clocking ratio multiplied by byte clock frequency, where the clocking ratio is calculated using the following formula:

$$(\text{NUM_LANES} * \text{PHY_DATA_WIDTH}) / (\text{PPC} * \text{BPP})$$
 whereby in this reference design NUM_LANES is 2, PHY_DATA_WIDTH is 8, PPC is 1 or 2 or 4, BPP is 10 (RAW10) with minimum ratio being 1, and byte clock frequency is 592 Mbps / 8 = 74 MHz.
 - NUM_LANES 2, PPC 1: $(2*8) / (1*10) = 1.6x$ Clocking Ratio; Clocking Required: $1.6 \times 74 \text{ MHz} = 118.4 \text{ MHz}$.
 - NUM_LANES 2, PPC 2: $(2*8) / (2*10) = 0.8x$ Clocking Ratio; Clocking Required: $1 \times 74 \text{ MHz} = 74 \text{ MHz}$.
 - NUM_LANES 2, PPC 4: $(2*8) / (4*10) = 0.4x$ Clocking Ratio; Clocking Required: $1 \times 74 \text{ MHz} = 74 \text{ MHz}$.
 For more information, refer to the [MIPI CSI/DSI IP User Guide \(FPGA-IPUG-02321\)](#).

7.1. Change the Camera Resolution

This section explains how to change the camera video resolution, which you can configure in PotPlayer.

When a resolution change is requested, the host sends the resolution configuration to the on-board embedded firmware over USB using the UVC protocol. The firmware performs sensor reconfiguration only when the camera is idle and no active video stream is running, ensuring stable and uninterrupted operation.

7.1.1. Selecting the 1080p Output Resolution

1. Open the PotPlayer application.
2. To configure output resolution, either press the shortcut Alt+D or click **PotPlayer > Open > Device Settings...**, as shown in [Figure 7.1](#).

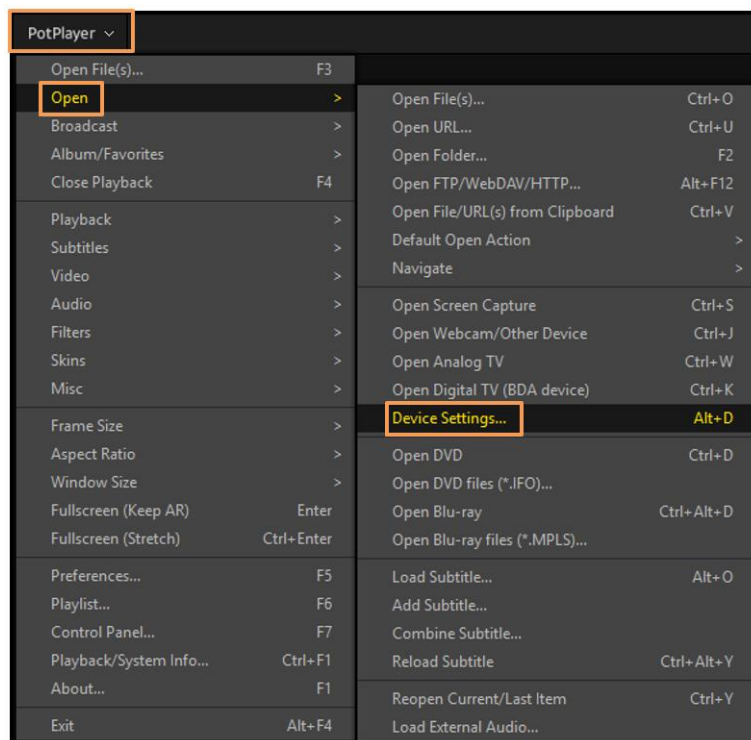


Figure 7.1. Device Settings in PotPlayer 1

3. In the Format drop-down menu shown in Figure 7.2, select the **YUY2 1920x1080** option. Then click **OK**.

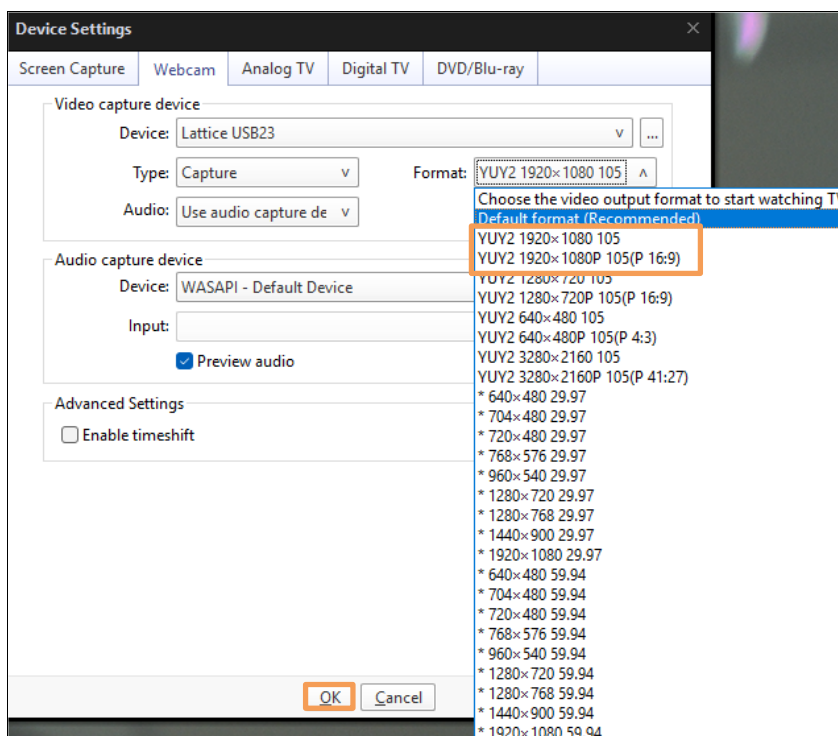


Figure 7.2. 1080p Selection in PotPlayer

4. Open video per normal settings, as described in the [Demonstration](#) section.

- (Optional) To adjust exposure, re-open **Device Settings**, as shown in the previous steps. Then, click the triple-dot menu to show the **Camera Control** menu (Figure 7.3).

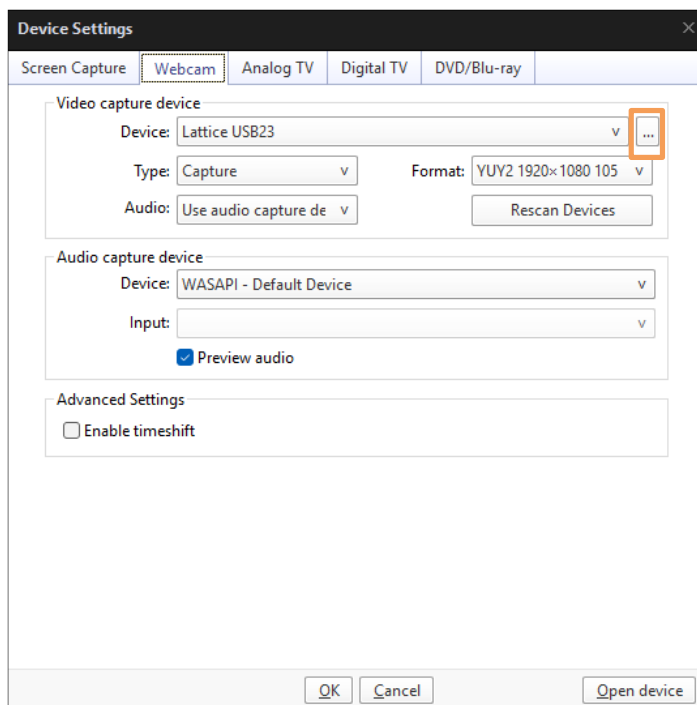


Figure 7.3. Camera Control Setting in PotPlayer 1

- (Optional) In the **Camera Control** menu, adjust exposure using the slider. Then, click **OK** to confirm (Figure 7.4).

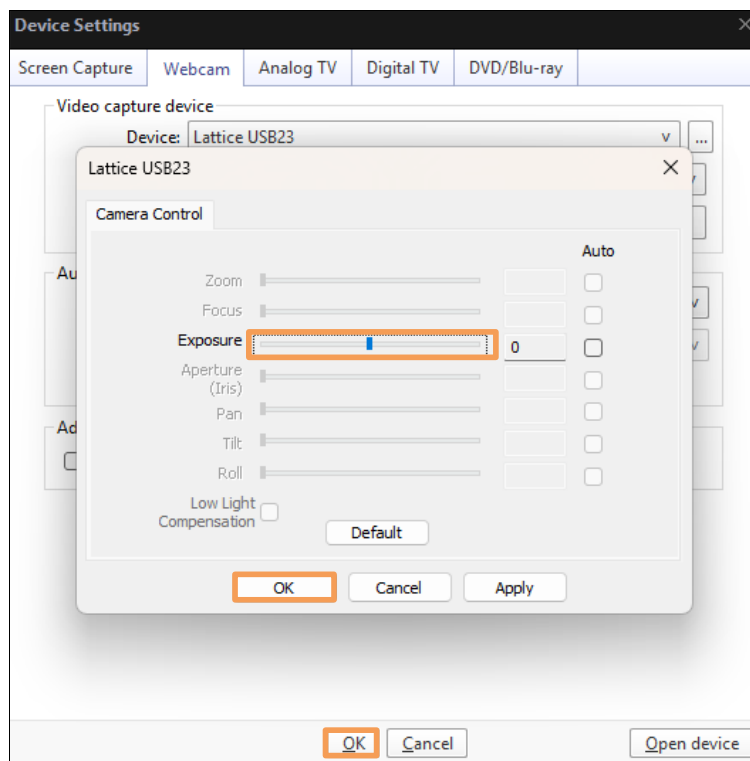


Figure 7.4. Camera Control Menu in PotPlayer 1

7.1.2. Selecting the 720p Output Resolution

1. Open the PotPlayer application.
2. To configure output resolution, either press the shortcut Alt+D or click **PotPlayer > Open > Device Settings...**, as shown in Figure 7.5.

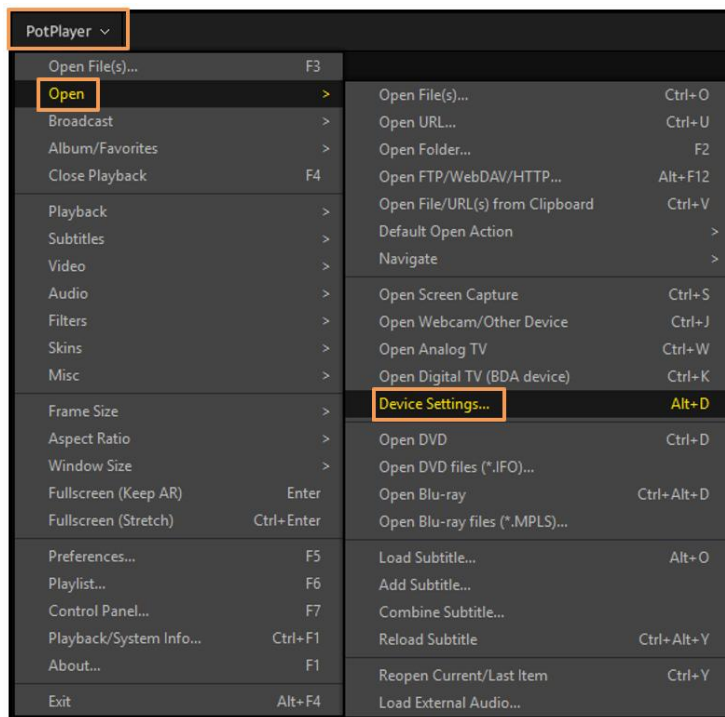


Figure 7.5. Device Settings in PotPlayer 2

3. In the Format drop-down menu shown in Figure 7.6, select the **YUY2 1280x720** option. Then click **OK**.

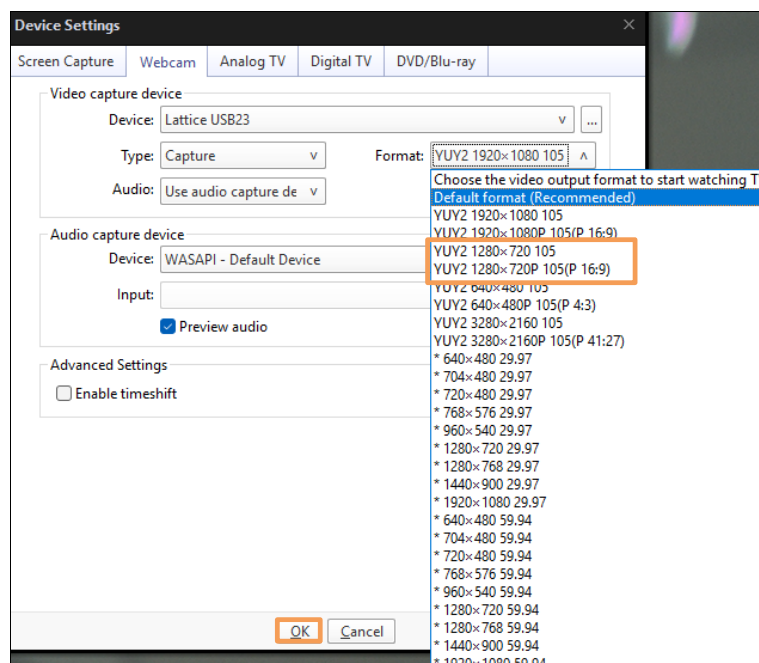


Figure 7.6. 720p Selection in PotPlayer

4. Open video per normal settings, as described in the [Demonstration](#) section.
5. (Optional) To adjust exposure, re-open **Device Settings**, as shown in the previous steps. Then click the triple-dot menu to show the **Camera Control** menu ([Figure 7.7](#)).

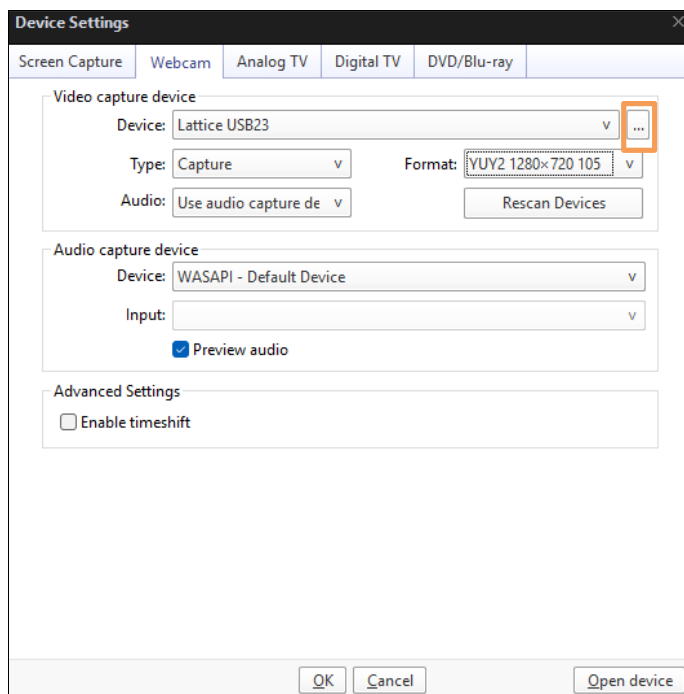


Figure 7.7. Camera Control Setting in PotPlayer 2

6. (Optional) In the **Camera Control** menu, adjust exposure using the slider. Then, click **OK** to confirm ([Figure 7.8](#)).

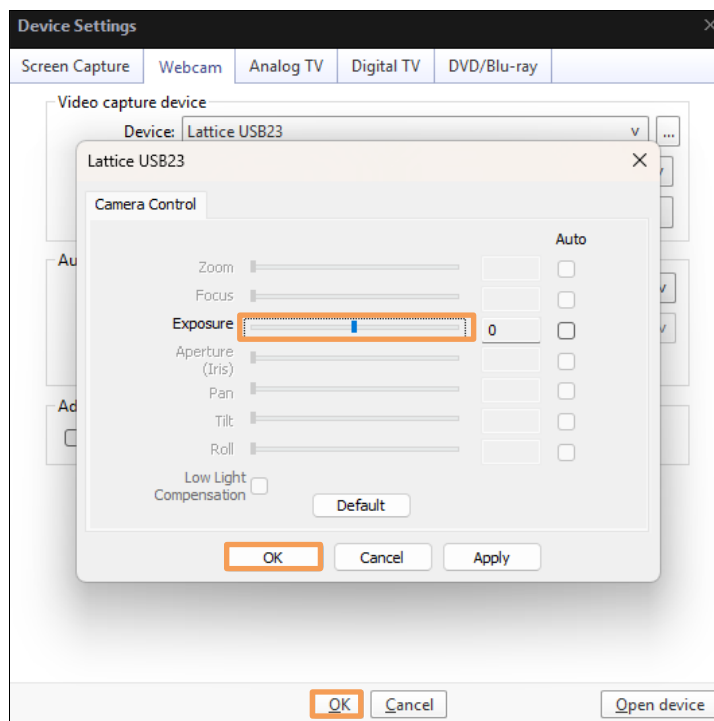


Figure 7.8. Camera Control Menu in PotPlayer 2

7.1.3. Selecting the 480p Output Resolution

1. Open the PotPlayer application.
2. To configure output resolution, either press the shortcut Alt+D or click **PotPlayer > Open > Device Settings...**, as shown in Figure 7.9.

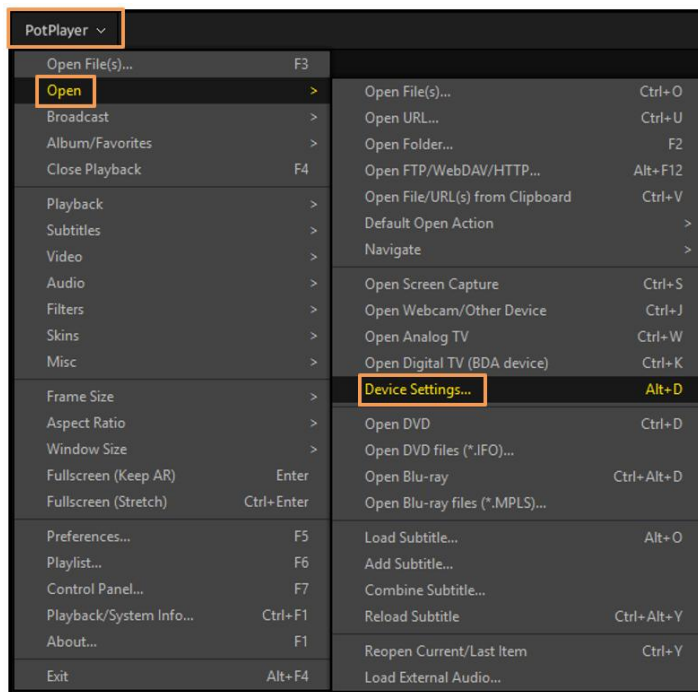


Figure 7.9. Device Settings in PotPlayer 3

3. In the Format drop-down menu shown in Figure 7.10, select the YUY2 640x480 option. Then click **OK**.

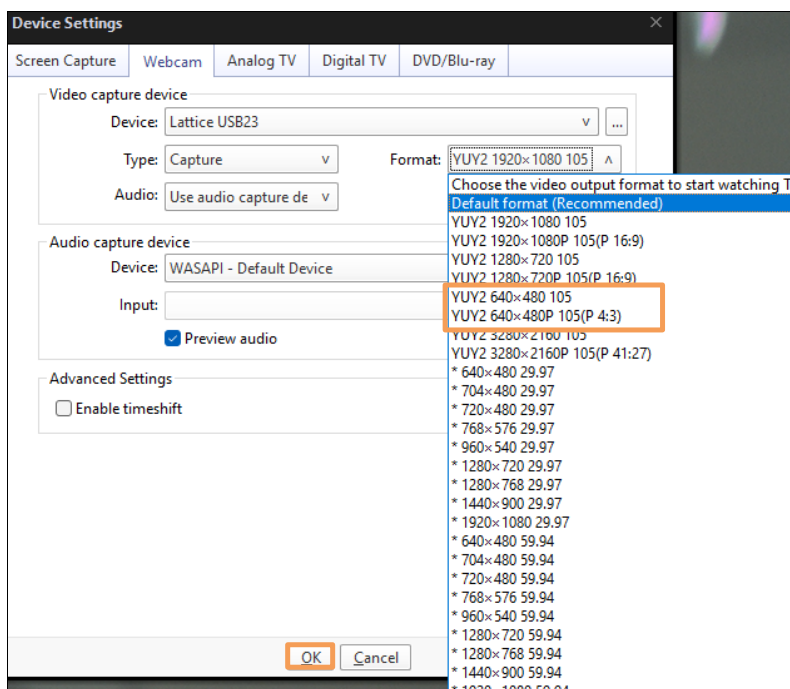


Figure 7.10. 480p Selection in PotPlayer

4. Open video per normal settings, as described in the [Demonstration](#) section.
5. (Optional) To adjust exposure, re-open **Device Settings**, as shown in the previous steps. Then, click the triple-dot menu to show the **Camera Control** menu ([Figure 7.11](#)).

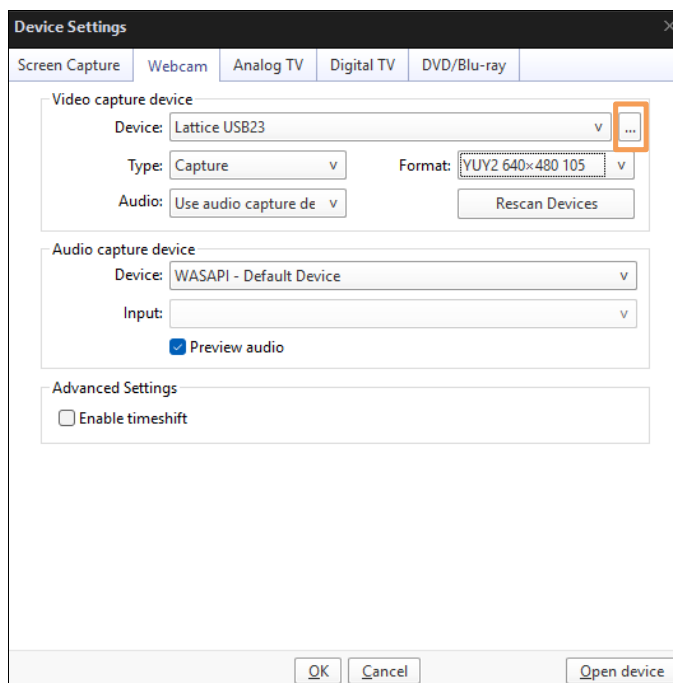


Figure 7.11. Camera Control Setting in PotPlayer 3

6. (Optional) In the **Camera Control** menu, adjust exposure using the slider. Then, click **OK** to confirm ([Figure 7.12](#)).

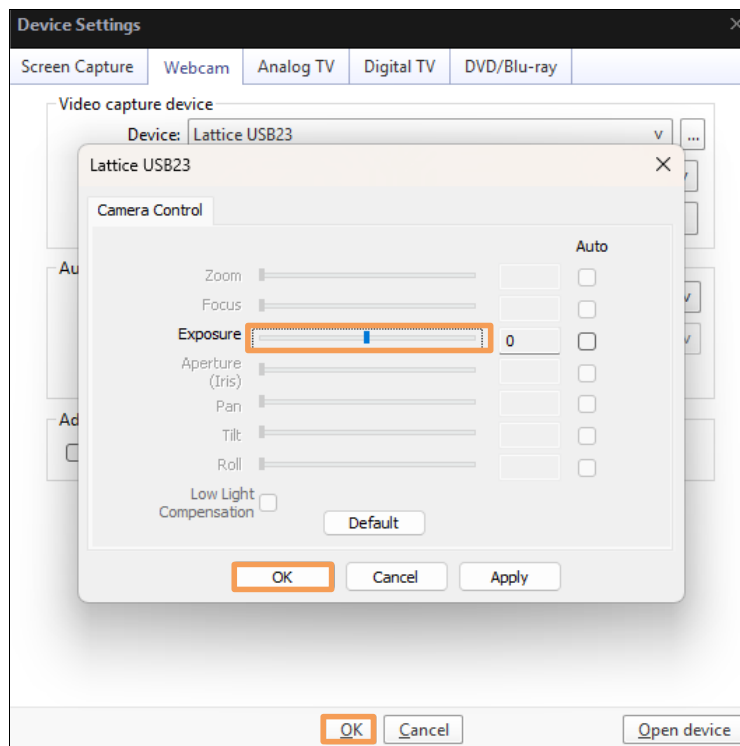


Figure 7.12. Camera Control Menu in PotPlayer 3

7.1.4. Selecting the 2160p Output Resolution (USB 3.0 Only)

1. Open the PotPlayer application.
2. To configure output resolution, either press the shortcut Alt+D or click **PotPlayer > Open > Device Settings...**, as shown in Figure 7.13.

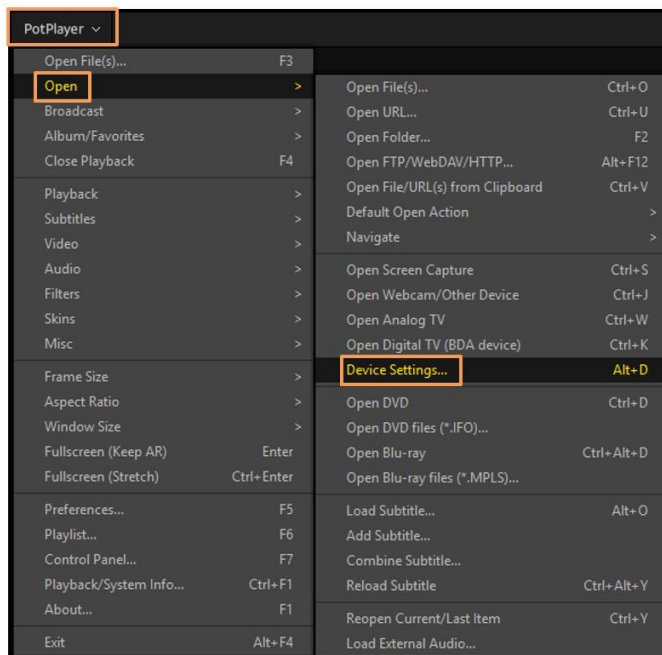


Figure 7.13. Device Settings in PotPlayer 4

3. In the Format drop-down menu shown in Figure 7.14, select the **YUY2 3280x2160** option. Then click **OK**.

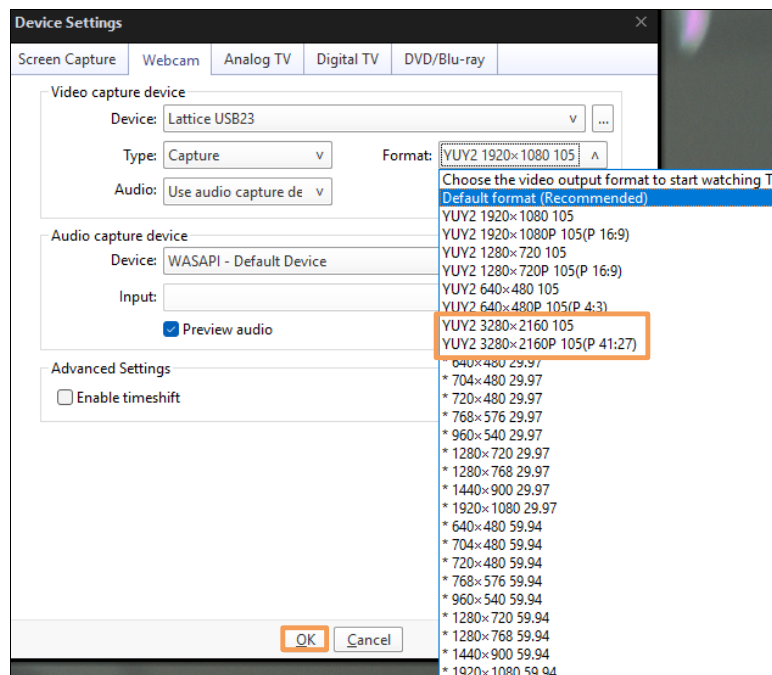


Figure 7.14. 2160p Selection in PotPlayer

4. Open video per normal settings, as described in the [Demonstration](#) section.

- (Optional) To adjust exposure, re-open **Device Settings**, as shown in the previous steps. Then, click the triple-dot menu to show the **Camera Control** menu (Figure 7.15).

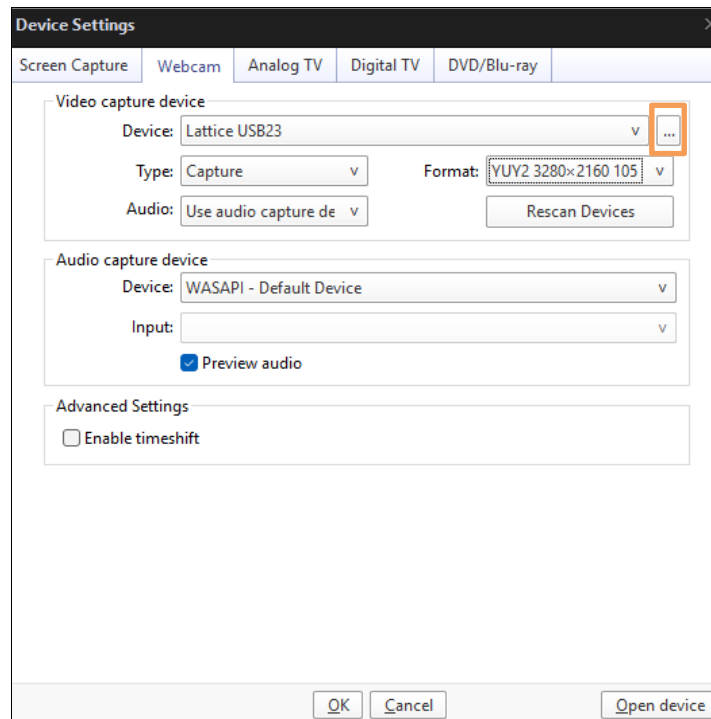


Figure 7.15. Camera Control Setting in PotPlayer 4

- (Optional) In the **Camera Control** menu, adjust exposure using the slider. Then, click **OK** to confirm (Figure 7.16).

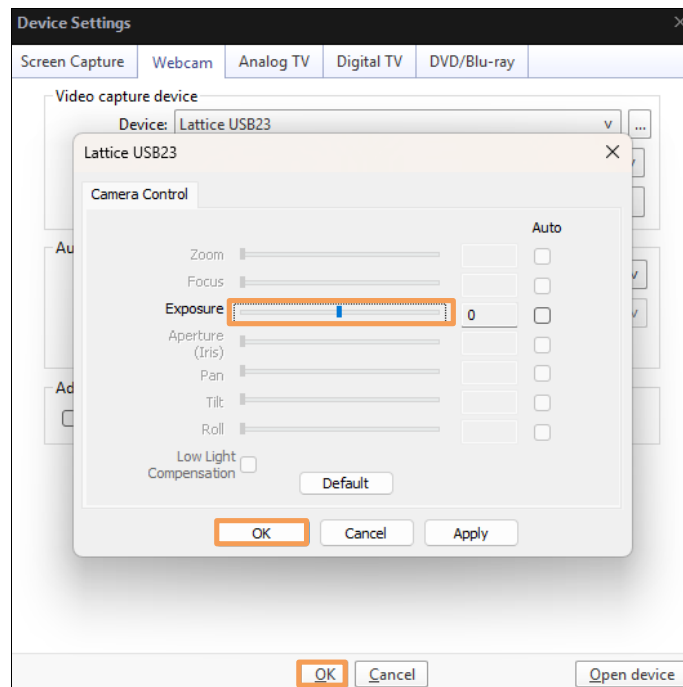


Figure 7.16. Camera Control Menu in PotPlayer 4

7.2. Changing IP Configuration to 1 PPC

This section describes how to configure the IP blocks for 1 PPC operation.

The reference design supports both 2 PPC and 1 PPC modes, allowing users to choose between higher maximum throughput (2 PPC) and reduced device resource utilization (1 PPC), as shown in [Table 8.1](#).

Note: The constraints file (.pdc) remains unchanged for this configuration.

7.2.1. Configuring the MIPI CSI/DSI IP

To configure the MIPI CSI/DSI IP, follow these steps:

1. Double-click the `mipi_csi_rx_inst` block to open the Module/IP Block Wizard.
2. Under the **Video Format** setting, set the **Number of Pixel per Clock (PPC)** to 1 ([Figure 7.17](#)).

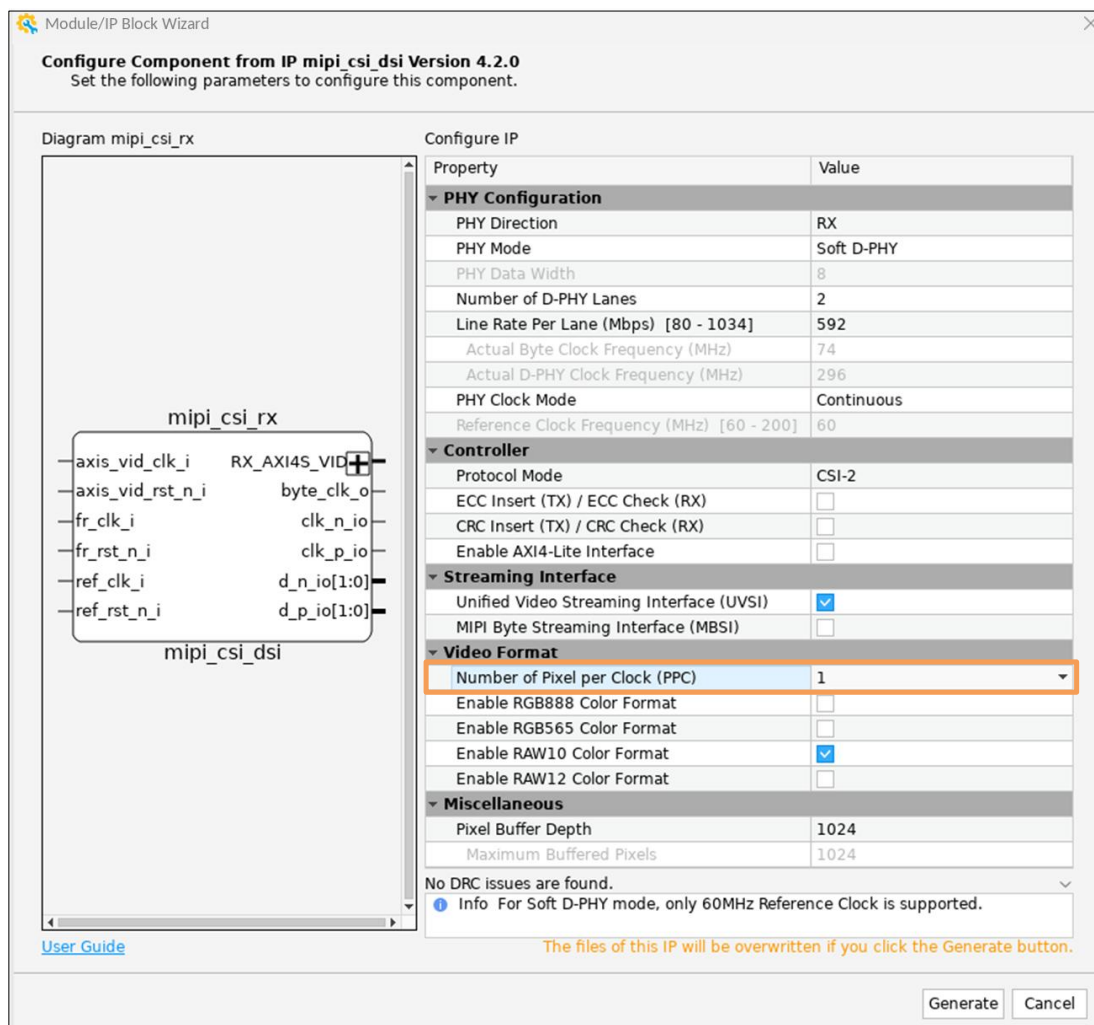


Figure 7.17. MIPI CSI/DSI Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window ([Figure 7.18](#)).
5. The unified video streaming data width has been updated, which causes existing UVSI connections to be disconnected. The connections can be re-established when `isp_debayer_inst` is reconfigured to the same PPC.

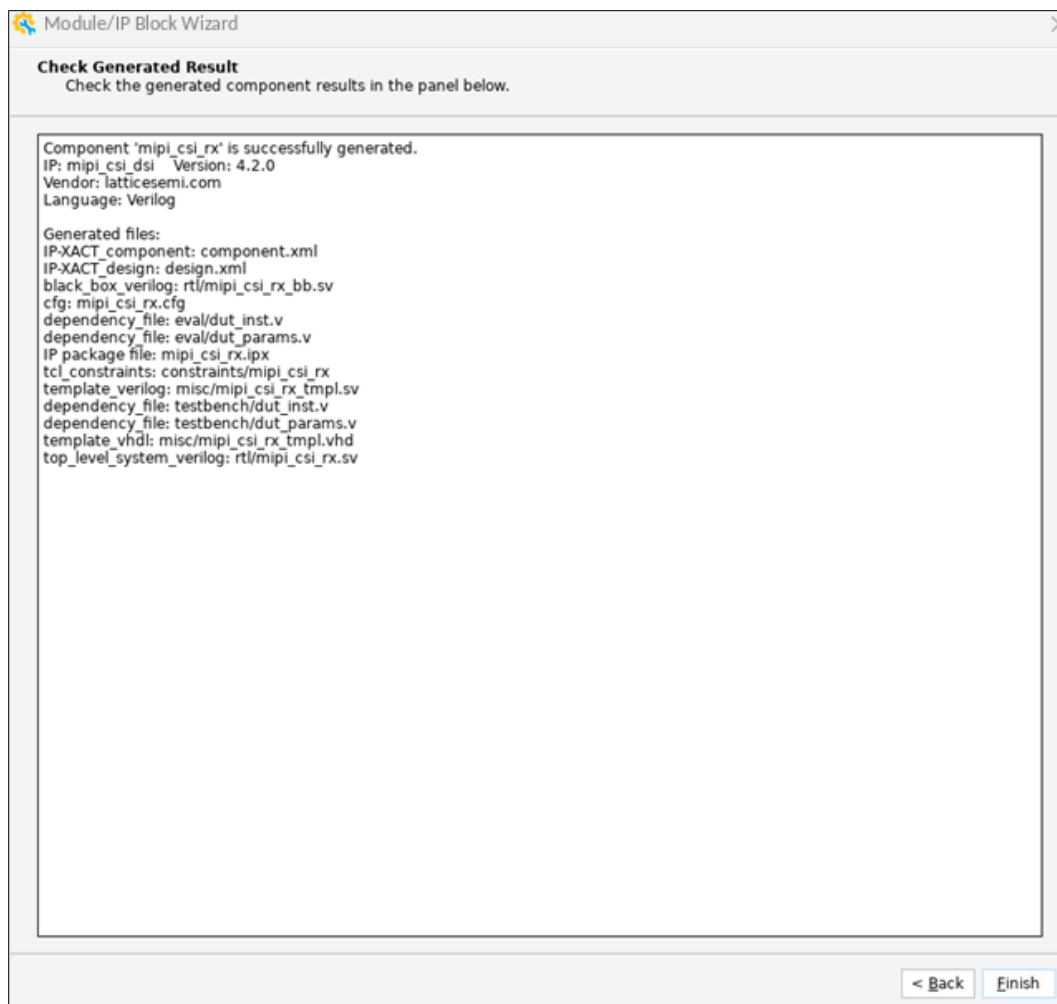


Figure 7.18. MIPI CSI/DSI IP Check Generated Result Window

7.2.2. Configuring the Debayer IP

To configure the Debayer IP, follow these steps:

1. Double-click the `isp_debayer_inst` block to open the Module/IP Block Wizard.
2. Under the **DEBAYER** setting, set the **Pixels per clock** to 1 (Figure 7.19).

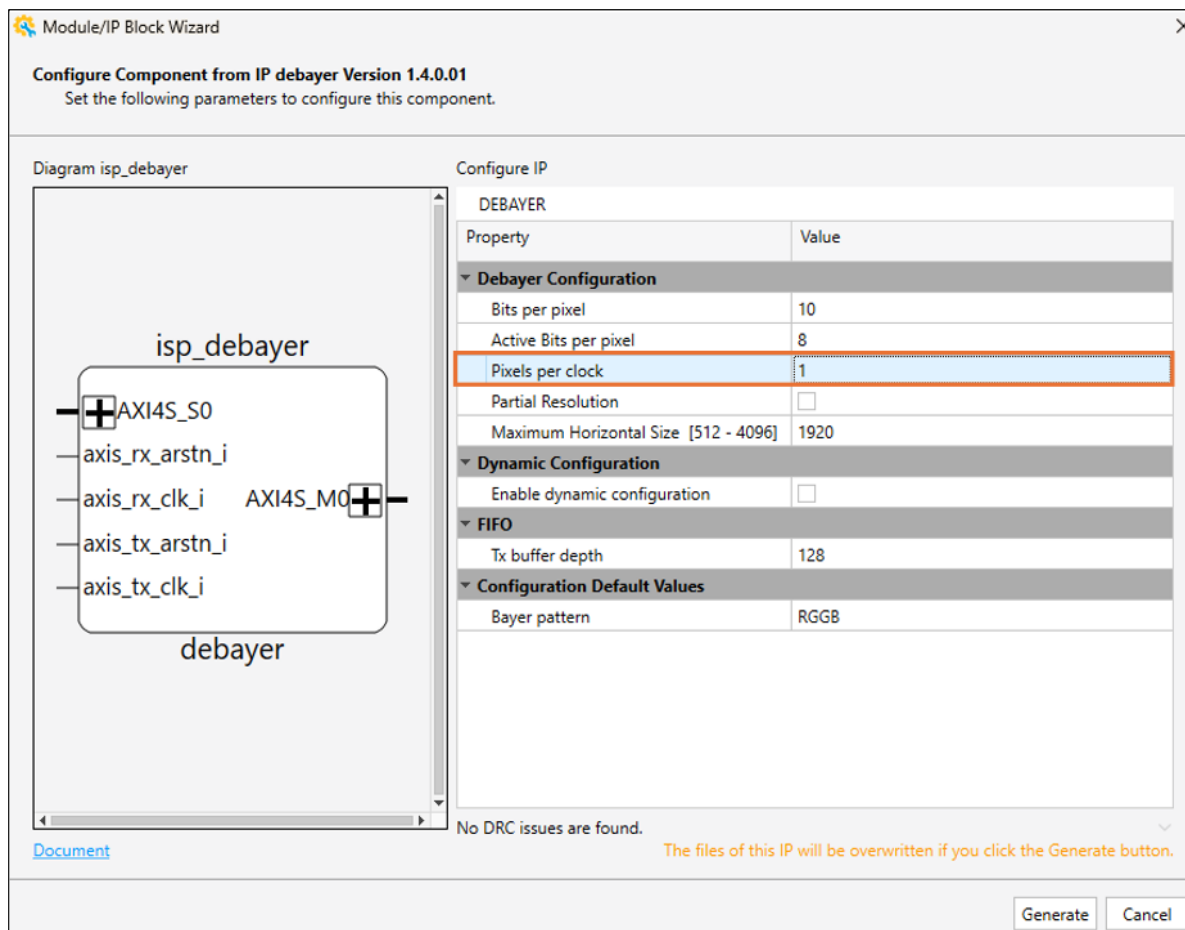


Figure 7.19. Debayer Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window (Figure 7.20).

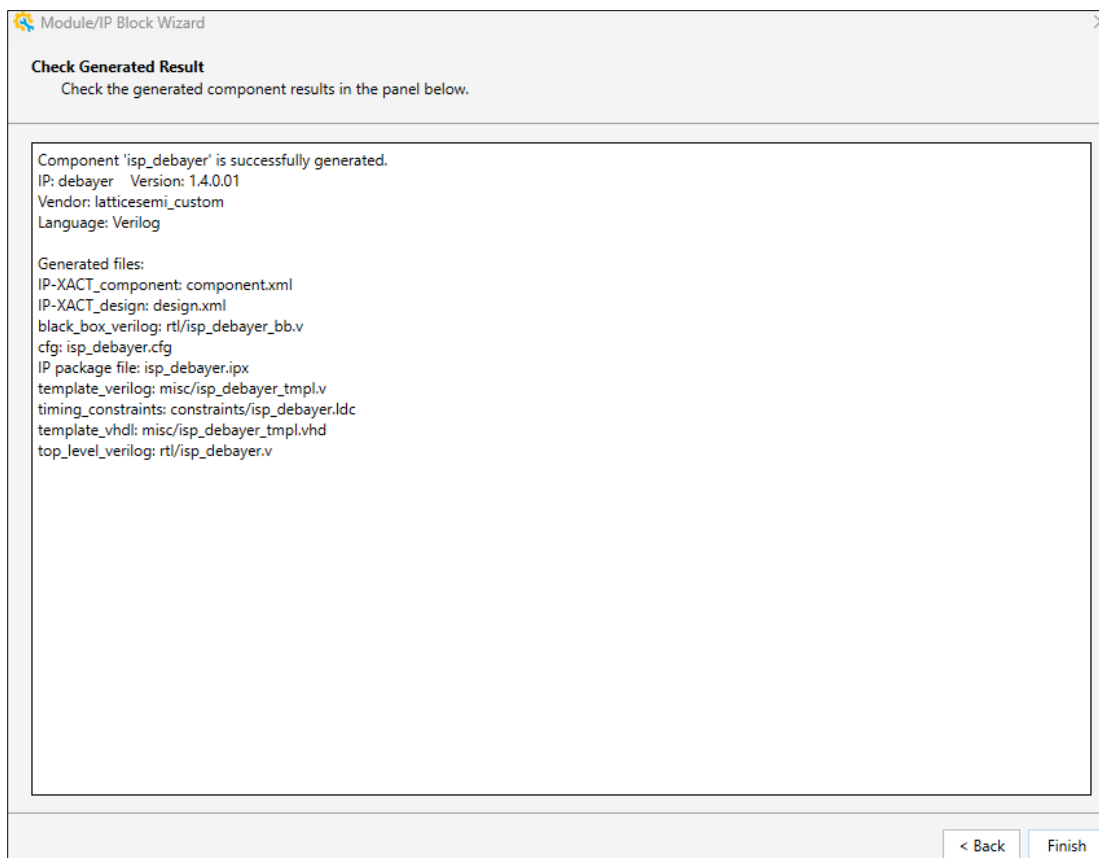


Figure 7.20. Debayer IP Check Generated Result Window

- The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect `isp_debayer_inst` `AXI4S_S0` to `mipi_csi_rx_inst` `AXI4S_VID`, as shown in [Figure 7.21](#).

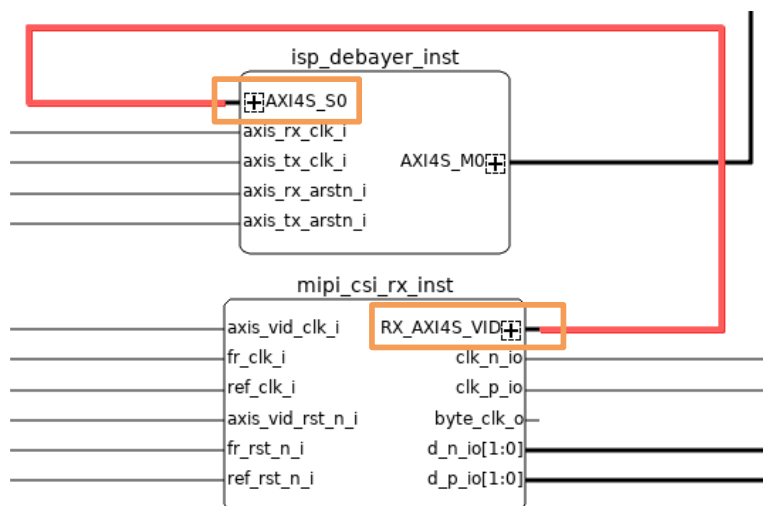


Figure 7.21. Propel Builder Connection Update for UVSI

7.2.3. Configuring the Color Correction Matrix IP

To configure the Color Correction Matrix IP, follow these steps:

1. Double-click the isp_ccm_inst block to open the Module/IP Block Wizard.
2. Under the **CCM** setting, set the **Pixels per clock** to 1 (Figure 7.22).

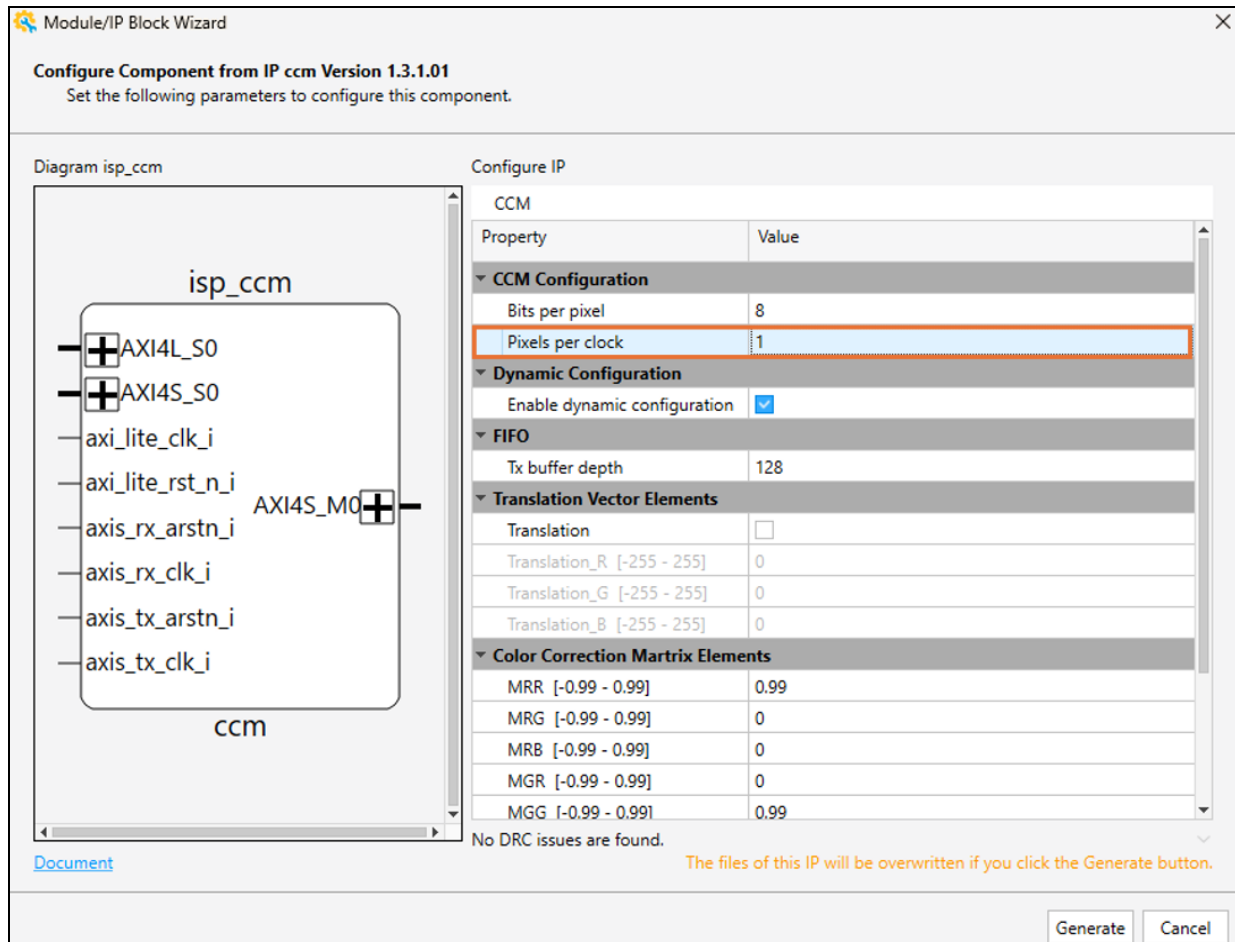


Figure 7.22. Color Correction Matrix Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window (Figure 7.23).

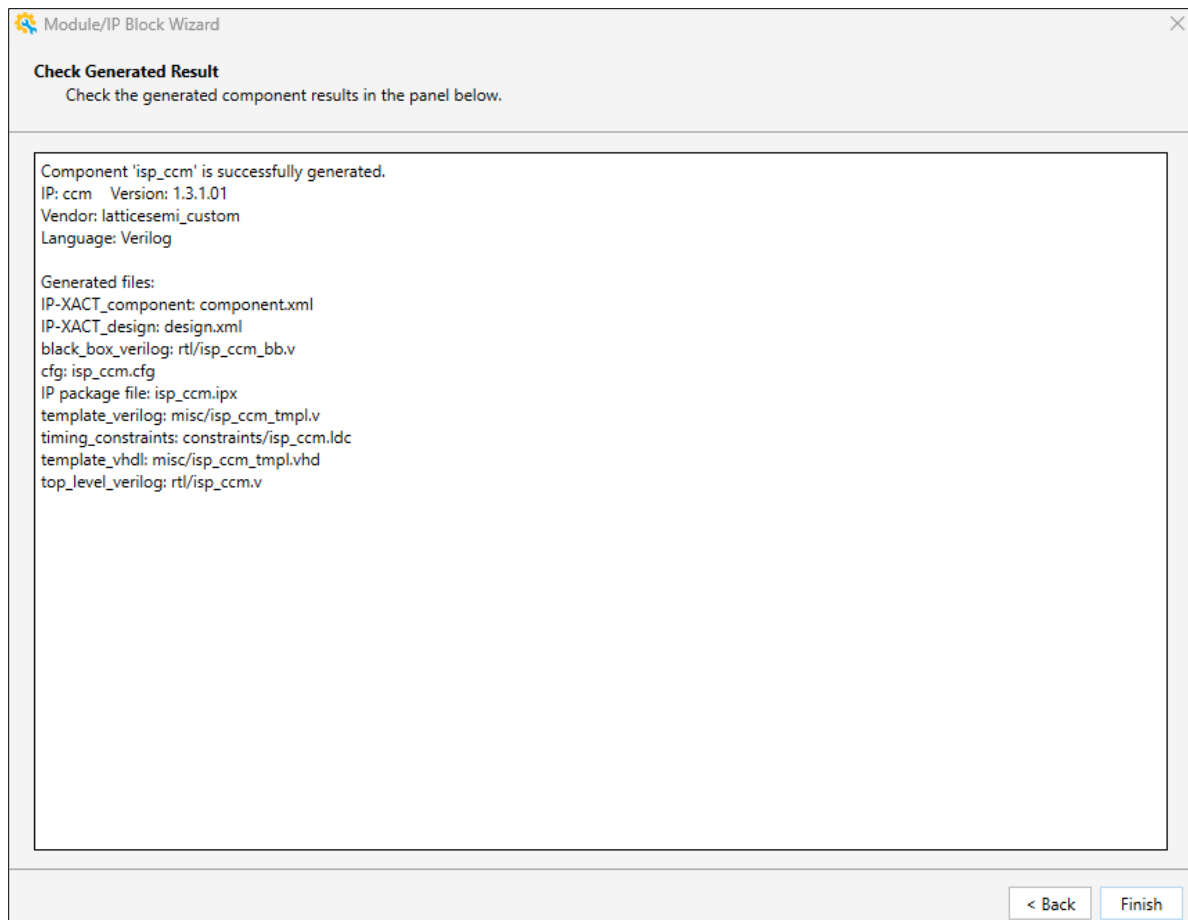


Figure 7.23. Color Correction Matrix IP Check Generated Result Window

5. The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect `isp_ccm_inst AXI4S_S0` to `isp_debayer_inst AXI4S_M0`. This is the same procedure used to connect `isp_debayer_inst` with `mipi_csi_rx_inst` through UVSI.

7.2.4. Configuring the Color Space Converter IP

To configure the Color Space Converter IP, follow these steps:

1. Double-click the `isp_csc_inst` block to open the Module/IP Block Wizard.
2. Under the **Input setting**, set the **Pixels per clock** to 1 (Figure 7.24).

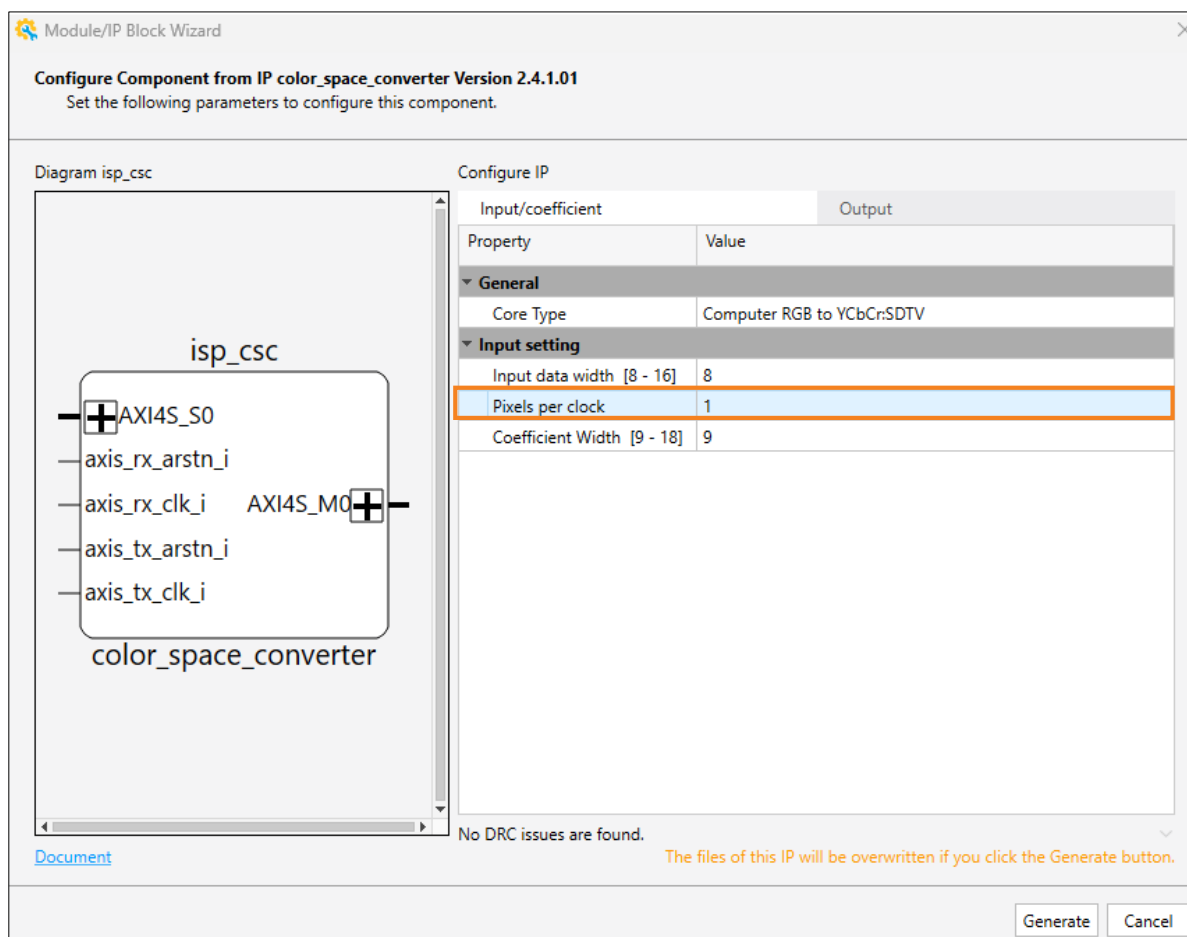


Figure 7.24. Color Space Converter Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window (Figure 7.25).

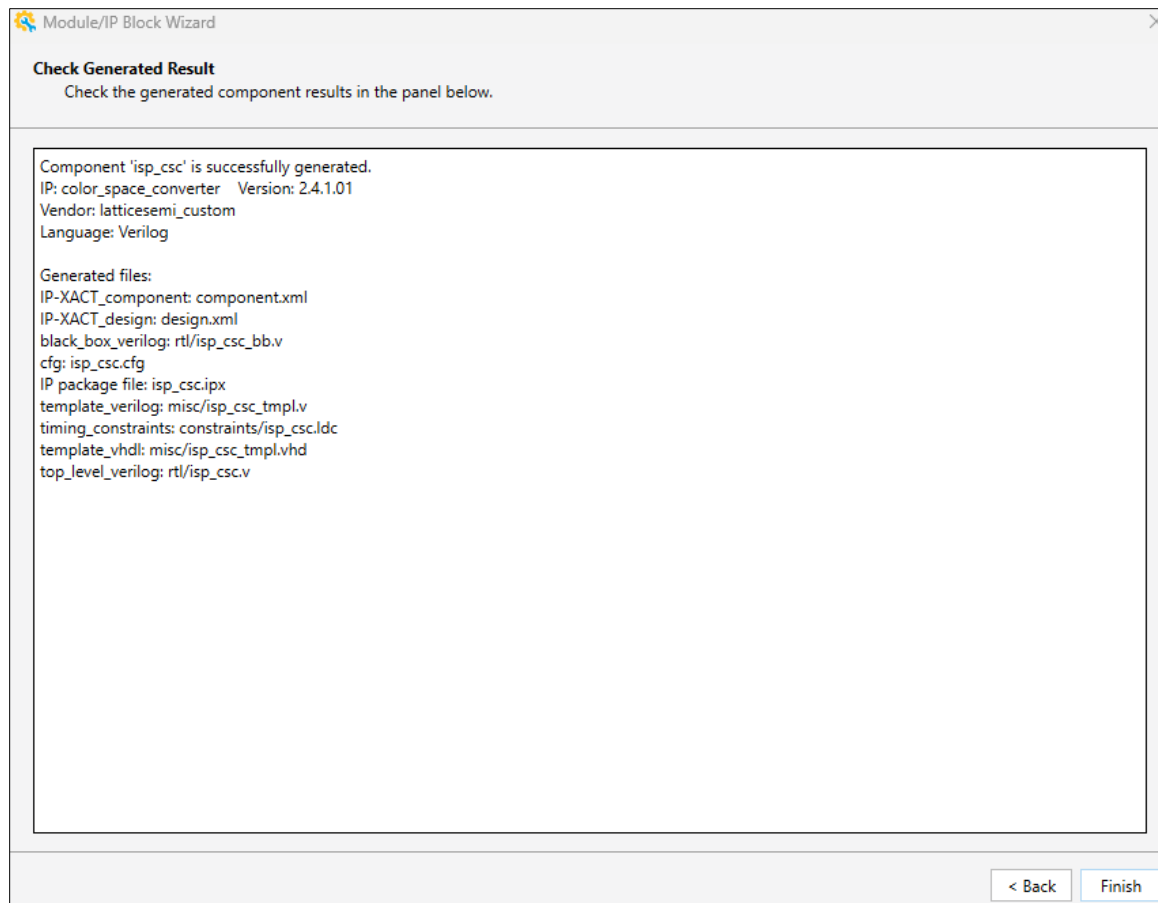


Figure 7.25. Color Space Converter IP Check Generated Result Window

5. The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect isp_csc_inst AXI4S_S0 to isp_ccm_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with mipi_csi_rx_inst through UVSI.

7.2.5. Configuring the Chroma Resampler IP

To configure the Chroma Resampler IP, follow these steps:

1. Double-click the isp_chroma_resampler_inst block to open the Module/IP Block Wizard.
2. Under the **Input setting**, set the **Pixels per clock** to 1 (Figure 7.26).

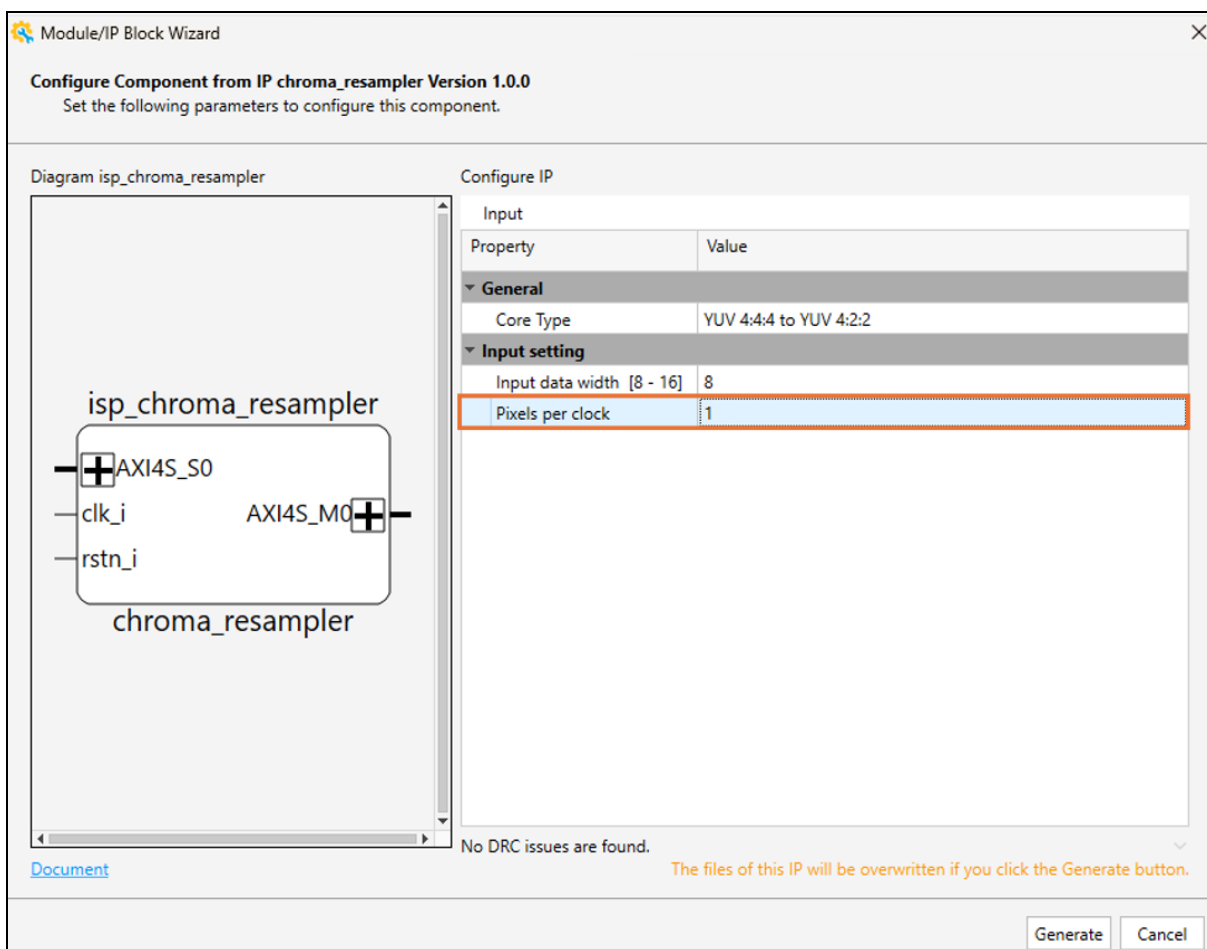


Figure 7.26. Chroma Resampler Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window (Figure 7.27).

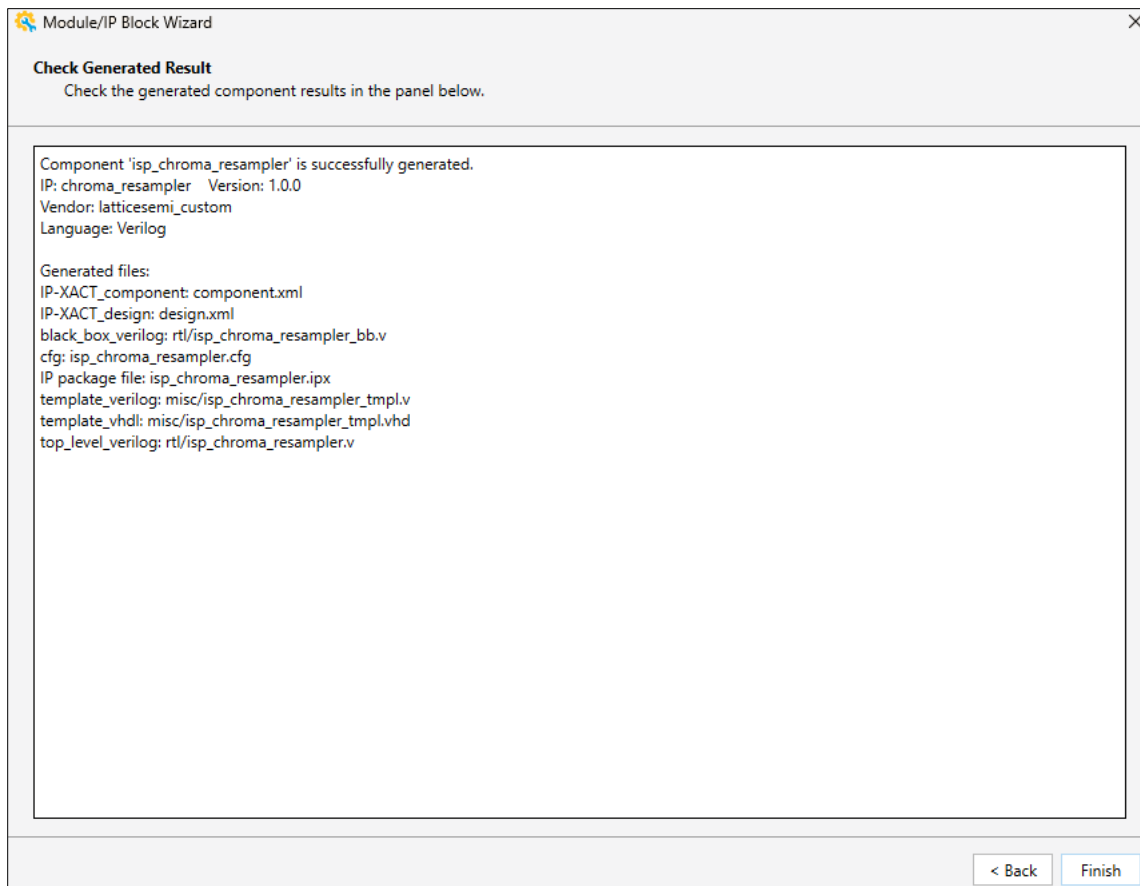


Figure 7.27. Chroma Resampler IP Check Generated Result Window

5. The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect `isp_chroma_resampler_inst AXI4S_S0` to `isp_csc_inst AXI4S_M0`. This is the same procedure used to connect `isp_debayer_inst` with `mipi_csi_rx_inst` through UVSI.

7.2.6. Configuring the Video to USB Bridge

To configure the Video to USB Bridge, follow these steps:

1. Double-click the `unified_video_to_usb_inst` block to open the Module/IP Block Wizard.
2. Under the **Input setting**, set the **Pixels per clock** to 1 (Figure 7.28).

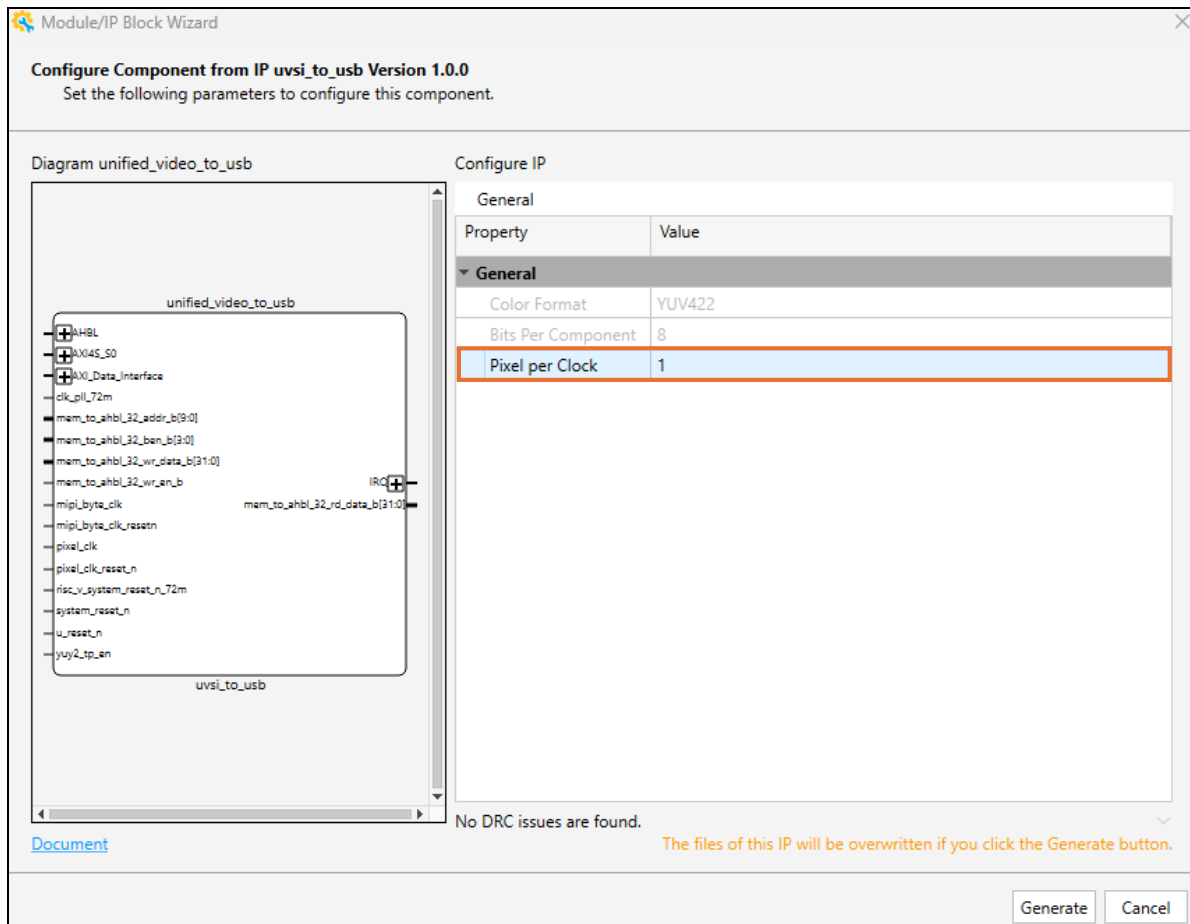


Figure 7.28. Video to USB Configuration

3. Click **Generate**.
4. After the generation is completed, click **Finish** in the **Check Generated Result** window (Figure 7.29).

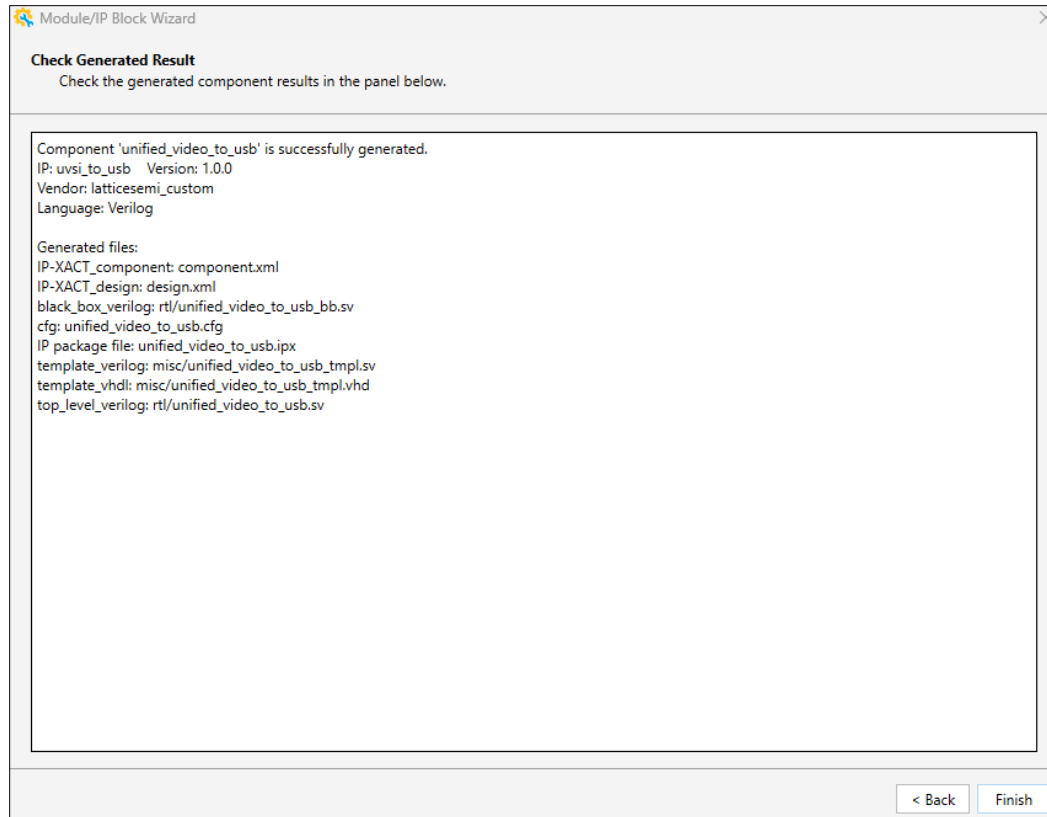


Figure 7.29. Video to USB Check Generated Result Window

5. The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect unified_video_to_usb_inst AXI4S_S0 to isp_chroma_resampler_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with mipi_csi_rx_inst through UVSI.

After reconfiguration is completed, refer to the [Running the Reference Design](#) section to generate the memory image and bitstream. After the bitstream is generated, refer to the [Programming the FPGA](#) section to program the bitstream into the FPGA.

7.3. Steps to Enable YUV422 Test Pattern

1. Go to <Project Directory>/propelsdk/riscv_mc/src and open the main.c file.
2. Uncomment the following line, as shown in [Figure 7.30](#).

```
//Test pattern enable GPIO
// 0x1 - yuy2 tp en (YUV422 test pattern)
// lsc_8_write((TP_EN_GPIO_INST_BASE_ADDR + 0x4), 0x1);

enable_camera();
```

Figure 7.30. Firmware Code to Enable the YUV422 Test Pattern

3. Write the data value 0x1 to enable the YUV422 test pattern.
4. Rebuild the Propel SDK project.
5. After the .mem file is generated, refer to Step 4 of the [Running Propel Builder and Radiant Project](#) section, to regenerate sysmem0_inst.

Note: YUV422 Test Pattern only supports 1080p resolution.

8. Resource Utilization

Table 8.1 below shows the example configurations validated on the CrossLinkU-NX Evaluation Board, including the resource utilization for the CrossLinkU-NX MIPI CSI-2 to USB UVC reference design. These configurations use the LIFCL-33U-9CTG104C device with Lattice Radiant software 2025.2.

Table 8.1. Resource Utilization for CrossLinkU-NX MIPI CSI-2 to USB UVC Reference Design

MIPI CSI/DSI		ISP (Debayer, CCM, CSC, Chroma Resampler)		Resource Utilization			
Lane Count	Line Rate (Mbps)	BPC	PPC	LUT4	Registers	DSP	EBR
x2	592	8	2	19403	12974	36	22
x2	592	8	1	15628	10755	21	22

9. Debugging

9.1. Errors when Updating the SPI Flash

- If there is a verification error when programming the .bit file, try changing the TCK frequency by setting **TCK Divider Setting** to greater than three. Click **Program** to restart programming (Figure 9.1).

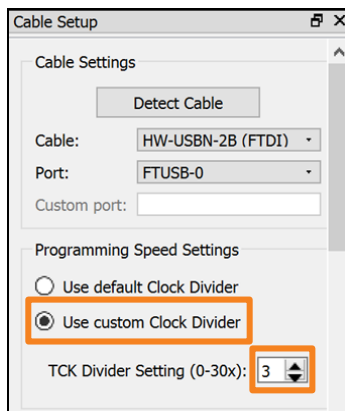


Figure 9.1. TCK Frequency Settings

- If the verification error issue still occurs, press and hold the **PROGRAMN** push button before clicking **Program**.
- If the device is not detected on Port: **FTUSB-0**, click **Detect Cable** and select Port: **FTUSB-1** (Figure 9.2).

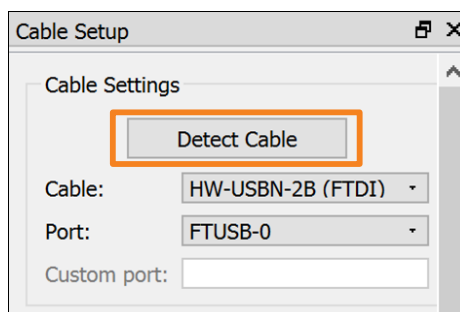


Figure 9.2. Port Selection

9.2. Timing Violations or Place and Route Failed

Timing violations may occur during compilation and may be caused by the Placement Iteration settings for Place and Route not being set optimally on the current machine. In addition, Place and Route may fail due to the signal being partially routed, as shown in Figure 9.3.

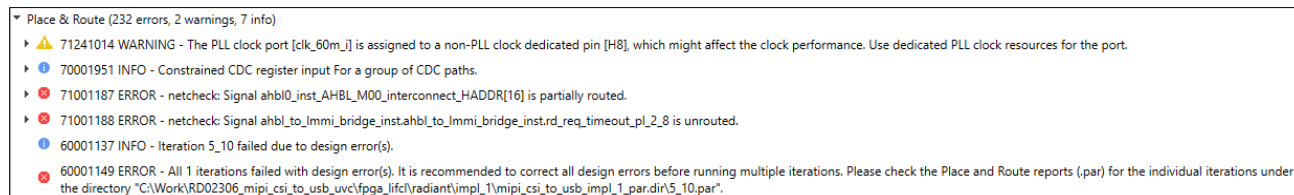


Figure 9.3. Place and Route Failed Message

To resolve these issues, follow these steps:

1. Go to **Project > Active Strategy > Place & Route Design Settings**.
2. In the settings window shown in [Figure 9.4](#), change both **Placement Iteration** and **Placement Save Best Run** from 1 to 10. Select **Stop Once Timing is Met**.

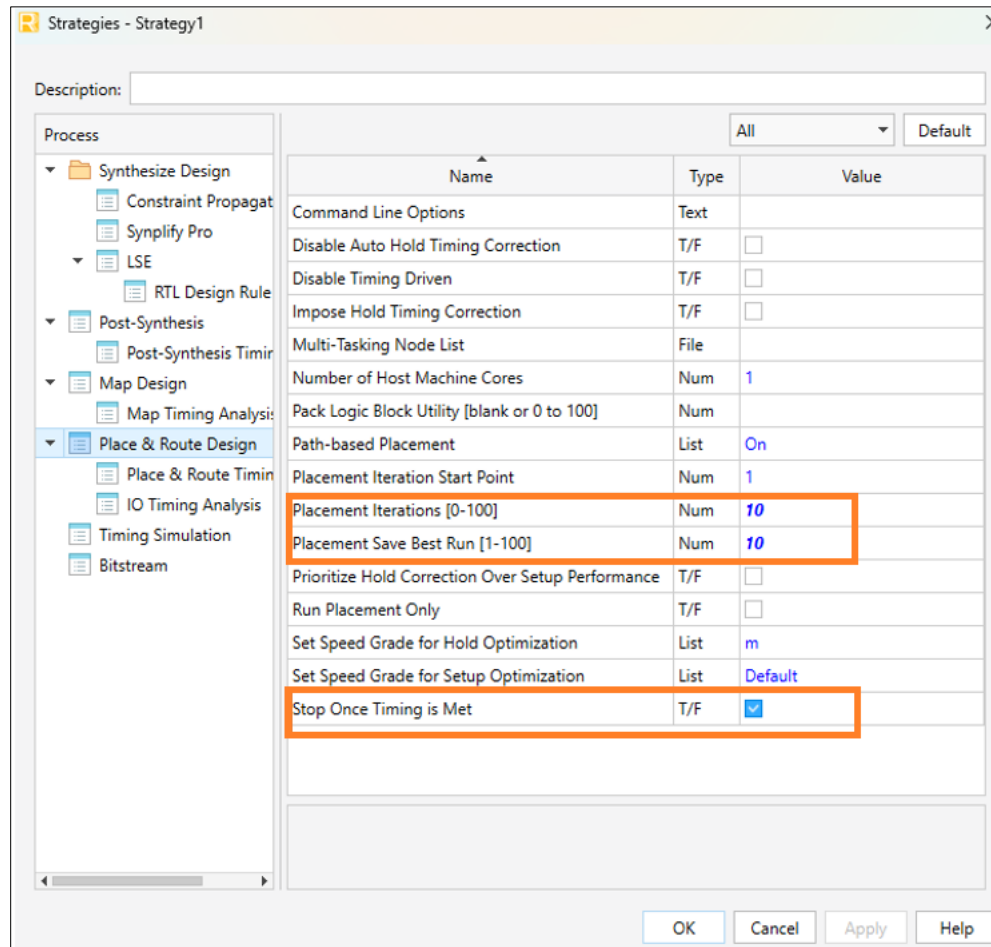


Figure 9.4. Place and Route Design Strategy Settings

3. Click **OK** to apply the changes.
4. Click **Export Files** to regenerate the bitstream.
5. If an iteration meets timing requirements, the Place and Route reports display the placement results.

9.3. No Display After Programming the Bitstream

If no display appears after programming the precompiled or recompiled bitstream, follow the steps below to enable the TPG and make sure that the hardware is functioning properly.

1. Program either the precompiled or recompiled bitstream onto the board.
2. Disconnect all cables to ensure that there is no power supplied to the CrossLinkU-NX Evaluation Board.
3. Remove the Raspberry Pi Camera Module V2 from the CrossLinkU-NX Evaluation Board.
4. Connect the CrossLinkU-NX Evaluation Board to the PC using a USB Type-C cable.
5. Repeat the steps described in the [Demonstration](#) section to run the reference design. Refer to [Selecting the 1080p Output Resolution](#) to configure the output resolution to 1080p, as this TPG only supports 1080p resolution.
6. The TPG video is displayed with a looping sequence of different colors.



Figure 9.5. TPG Video

9.4. Black Screen When Switching Resolution

The display may turn black when switching resolutions during active display (Figure 9.6). To avoid this issue, press the **Stop Display** button before changing the resolution. After the resolution switch is complete, press the **Start Display** button to resume displaying.

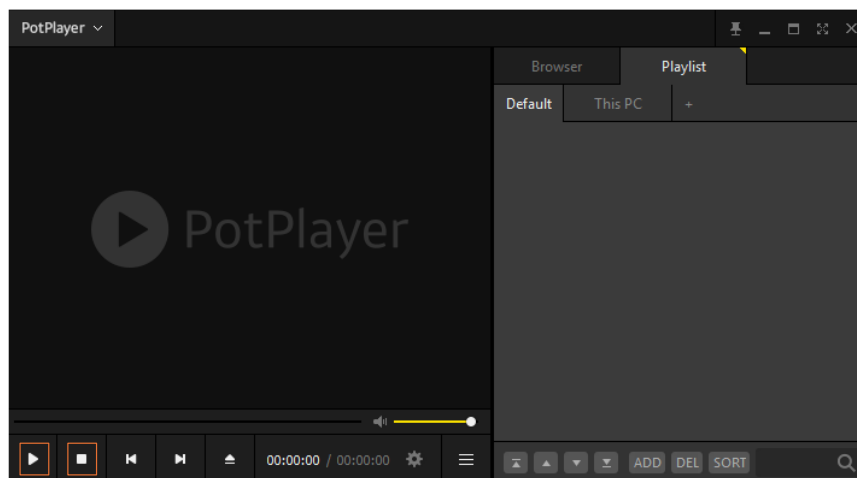


Figure 9.6. PotPlayer GUI

9.5. INFO and WARNING Messages in Lattice Propel Builder Software

9.5.1. Opening the Project File

If the following INFO and WARNING messages appear (Figure 9.7) when opening the .sbx file, the INFO messages indicate an interface port bus width mismatch between IP blocks. This is expected, as not all bus signals or bits are fully used in the design.

The WARNING indicates that the required IP is not installed in the <IP Installation Path>. Follow the steps in the [Running Propel Builder and Radiant Project](#) section to install the corresponding IPK files for the required IPs.

Figure 9.7. INFO and WARNING when Opening a Project File

When validating and generating the design, several INFO and WARNING messages ([Figure 9.8](#)) appear. These messages are expected and do not indicate errors. The WARNING messages indicate that some modules do not contain a bus interface memory map. This is expected behavior because these bus interfaces do not require memory mapping. During the generation process, some port connection warnings may also appear, indicating that certain ports remain unconnected. These ports are intentionally left unconnected, as they are not used in this design. Finally, the INFO messages indicate that there is no available driver for the listed bridge components as these bridge components do not require software drivers for the intended operation of this reference design.

Figure 9.8. INFO and WARNING when Validating and Generating a Design

References

- [CrossLinkU-NX Devices](#) web page
- [Lattice Radiant Timing Constraints Methodology \(FPGA-AN-02059\)](#)
- [USB 2.0/3.2 IP \(FPGA-IPUG-02237\)](#)
- [MIPI CSI/DSI IP User Guide \(FPGA-IPUG-02321\)](#)
- [Lattice Propel Builder Software](#) web page
- [Lattice Radiant Software](#) web page
- [Lattice Insights](#) for Lattice Semiconductor training courses and learning plans

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For frequently asked questions, please refer to the Lattice Answer Database at www.latticesemi.com/Support/AnswerDatabase.

Revision History

Revision 1.4, September 2026

Section	Change Summary
All	Updated the IP name <i>MIPI CSI/DSI RX IP</i> to <i>MIPI CSI/DSI IP</i> globally.
Abbreviations in This Document	Updated this section.
Introduction	- In Table 1.1. Summary of the Reference Design: - updated the Lattice Propel SDK tool version from 2025.2 to 2026.1; - updated the Lattice Propel Builder tool version from 2025.2 to 2026.1; - updated the Lattice Radiant software tool version from 2025.2 to 2026.1; - updated the Lattice Radiant Programmer tool version from 2025.2 to 2026.1; - changed the IP Version entry <i>MIPI CSI/DSI RX IP Core version 1.0.0.02 (Custom IP)</i> to <i>MIPI CSI/DSI IP Core version 4.2.0</i>; - updated the Color Space Converter IP version from 2.4.1.01 to 2.4.1.03; - updated the Color Correction Matrix IP version from 1.3.1.01 to 1.3.1.03; - updated the Debayer IP version from 1.4.0.01 to 1.4.0.03; - updated the USB23 IP version from 3.1.1 to 3.2.0; - removed <i>FIFO_DC IP version 2.3.0</i> from the IP Version row. - In the Features section: - removed the IP documentation reference path <i><IP Installation Path>/latticesemi_custom_ip_mipi_csi_dsi_1.0.0.02/doc/IP_Design_Notes.pdf</i>; - updated the IP documentation reference paths of the Debayer, Color Correction Matrix, and Color Space Converter IP Cores to reflect the corresponding updated IP versions.
Directory Structure and Files	Updated Figure 2.1. Directory Structure .
Functional Description	- Updated Figure 3.1. Reference Design Block Diagram. - In the Design Components section, changed the statement <i>The ISPs operate with a configuration of 8 bits per component (BPC) and 2 pixels per clock (PPC), with a maximum supported video resolution of 1080p30</i> to <i>The ISPs operate with a configuration of 8 bits per component (BPC) and 2 pixels per clock (PPC) with a maximum supported video resolution of 3280x2160</i>. - In the Unified Interconnect section: - updated the section title from <i>AHB-Lite Interconnect</i> to the current; - changed the statement <i>This IP provides the communicating interface between RISC-V soft processor and peripherals that support AHB-Lite interface, such as System Memory and AHB-Lite to LMMI Bridge</i> to <i>This IP provides the communicating interface between RISC-V soft processor and peripherals that support AHB-Lite and APB interface, such as System Memory, UART, and I2C Controller</i>. - Removed the old APB Interconnect and AHB-Lite to APB Converter sub-sections. - In the MIPI CSI/DSI IP Core section: - updated the section title from <i>MIPI CSI/DSI RX IP Core</i> to the current; - removed the statement <i>For complete specifications, including interface signals and parameter details, refer to the MIPI CSI/DSI RX IP documentation: <IP Installation Path>/latticesemi_custom_ip_mipi_csi_dsi_1.0.0.02/doc/IP_Design_Notes.pdf</i>. - In the Debayer IP Core, Color Correction Matrix IP Core, and Color Space Converter IP Core sections, updated the IP documentation reference paths to reflect the corresponding updated IP versions. - In Table 3.1. Video to USB Bridge Parameters, changed the Default value of <code>PIXEL_PER_CLK</code> from 4 to 2. - Updated Figure 3.2. Clocking Scheme Overview. - Updated Figure 3.3. Reset Scheme Overview.
Running the Reference Design	Added the following note to the introductory paragraph: <i>The screenshots provided are for reference only. Details may vary depending on the</i>

Section	Change Summary
	<i>version of the IP, reference design, or software being used. If there have been no significant changes to the GUI, a screenshot may reflect an earlier version of the IP and reference design.</i> - In the Running Propel SDK Project section: - changed the path separators of the workspace folder path in Step 2, the projects root directory path in Step 5, and the riscv_mc.mem output path in Step 9 from backslashes to forward slashes. - In the Running Propel Builder and Radiant Project section: - changed the path separators of the IPK installation paths from backslashes to forward slashes; - removed <Project Directory>/rd02306_mipi_csi_to_usb_uvc/misc/custom_ip/latticesemi_custom_mipi_csi_dsi_1.0.0.02.ipk from the list of IPKs to be installed; - updated the Color Correction Matrix, Color Space Converter, and Debayer IPK file names to reflect corresponding updated IP versions; - updated Figure 5.8. IP on Local Tab in IP Catalog, Figure 5.11. IPs Installed from Files, and Figure 5.13. Base Address Assignment.
Implementing the Reference Design on Board	Updated Figure 6.15. Video Output on PotPlayer and Figure 6.18. Example of Video Information .
Customizing the Reference Design	Table 7.1. IP Configuration for Supported Resolutions: - changed the <i>Resolution</i> column header to <i>Resolution/FPS</i>; - added the <i>USB Mode</i> column; - replaced the 1080p30 and 720p30 rows with seven rows covering 3280x2160, 1920x1080, 1280x720, and 640x480 for both 3.0 (SuperSpeed) and 2.0 (High-Speed) USB modes; - changed the table note statement from <i>For more information, refer to <Project Directory>/latticesemi_custom_ip_mipi_csi_dsi_1.0.0.02/doc/IP_Design_Notes.pdf for the IP user guide to</i> <i>For more information, refer to the MIPI CSI/DSI IP User Guide (FPGA-IPUG-02321).</i> - In the Selecting the 1080p Output Resolution section: - added two optional steps describing how to adjust exposure through the Camera Control menu; - updated Figure 7.2. 1080p Selection in PotPlayer; - added Figure 7.3. Camera Control Setting in PotPlayer 1 and Figure 7.4. Camera Control Menu in PotPlayer 1. - In the Selecting the 720p Output Resolution section: - added two optional steps describing how to adjust exposure through the Camera Control menu; - updated Figure 7.6. 720p Selection in PotPlayer; - added Figure 7.7. Camera Control Setting in PotPlayer 2 and Figure 7.8. Camera Control Menu in PotPlayer 2. - Added the Selecting the 480p Output Resolution and Selecting the 2160p Output Resolution (USB 3.0 Only) sections. - In the Changing IP Configuration to 1 PPC section, added the following statement: <i>The reference design supports both 2 PPC and 1 PPC modes, allowing users to choose between higher maximum throughput (2 PPC) and reduced device resource utilization (1 PPC), as shown in Table 8.1. Resource Utilization for CrossLinkU-NX MIPI CSI-2 to USB UVC Reference Design.</i> - In the Configuring the MIPI CSI/DSI IP section: - updated the section title from <i>Configuring the MIPI RX IP</i> to the current; - in Step 2, changed <i>Under the Byte-to-Pixel setting</i> to <i>Under the Video Format setting</i>; - updated Figure 7.17. MIPI CSI/DSI Configuration and Figure 7.18. MIPI CSI/DSI IP Check Generated Result Window. - In the Configuring the Debayer IP section, updated Figure 7.21. Propel Builder Connection Update for UVS1.

Section	Change Summary
References	Updated this section.

Revision 1.3, July 2026

Section	Change Summary
All	- This revision applies to the MIPI CSI-2 to USB UVC Reference Design v1.1.0. - Updated the document title to the current.
Abbreviations in This Document	Updated this section.
Introduction	- In the Quick Facts section, added the following note: <i>In some instances, the IPs may be updated without changes to the corresponding user guides. The user guide of each IP may reflect an earlier IP version but remains fully compatible with the later IP version.</i> - In the Features section, changed the IP document reference path prefix from <Project Directory> to <IP Installation Path>.
Functional Description	- Changed the IP document reference path prefix from <Project Directory> to <IP Installation Path> in the MIPI CSI/DSI RX IP Core, Debayer IP Core, Color Correction Matrix IP Core, and Color Space Converter IP Core descriptions. - In the USB23 Controller section: changed the statement <i>the USB controller talks to the system Memory to the USB controller talks to the True Dual-Port (TDP) Memory.</i> - In the GPIO section: changed <i>This GPIO is used to select the data either from the camera sensor RX DPHY or MIPI RAW10 Test Pattern Generator, and Image Signal Processing stream or YUV422 Test Pattern Generator to This GPIO is used to select between the Image Signal Processing data stream and YUV422 Test Pattern Generator.</i> - In the Debayer IP Core section: changed the statement <i>The image sensor outputs pixel data in Bayer format to The Debayer IP accepts pixel data input in Bayer format from source such as image sensor for processing.</i> - In the Video to USB Bridge section: - added <i>This TPG only supports 1080p resolution;</i> - clarified that the TDP Memory offers dedicated storage that supports simultaneous read and write operations <i>between RISC-V CPU and USB23 Controller.</i> - Updated Figure 3.2. Clocking Scheme Overview.
Running the Reference Design	- In the introductory paragraph, added <i>If you want to evaluate the reference design using the precompiled bitstream, you can skip the build steps in this section and proceed directly to the Precompiled Bitstream section.</i> - In the Running Propel SDK Project section: - changed the workspace folder directory in Step 2 from <Project Directory>\RD02306_mipi_csi_to_usb_uvc\propelsdk to <Project Directory>\rd02306_mipi_csi_to_usb_uvc\fgpa_lifcl\propelsdk; - changed the root directory in Step 5 from <Project Directory>\RD02306_mipi_csi_to_usb_uvc\propelsdk\riscv_mc to <Project Directory>\rd02306_mipi_csi_to_usb_uvc\fgpa_lifcl\propelsdk\riscv_mc; - changed the riscv_mc.mem file directory in Step 9 from <Project Directory>\RD02306_mipi_csi_to_usb_uvc\propelsdk\riscv_mc\Debug\riscv_mc.mem to <Project Directory>\rd02306_mipi_csi_to_usb_uvc\fgpa_lifcl\propelsdk\riscv_mc\Debug\riscv_mc.mem. - In the Running Propel Builder and Radiant Project section: - removed the old Figure 5.8. Lattice Propel Builder Window and Figure 5.17. TCL Console Output after Generating the Design and related descriptions; - added the following note below Step 2, <i>The expected INFO messages and WARNING appear after opening the project file. Refer to the Debugging section for detailed explanation of these messages;</i> - updated Figure 5.11. Mapping the System Memory Initialization Files;

Section	Change Summary
	- removed the following statement from Step 6 and Step 7: <i>Ensure the TCL console shows no errors, and only the expected INFO and WARNING messages appear;</i> - added the following note below Step 7, <i>Ensure the TCL console shows no errors, and only the expected INFO and WARNING messages appear. Refer to the Debugging section for detailed explanation of these messages.</i>
Implementing the Reference Design on Board	- In the Programming the FPGA section: - added the Recompiled Bitstream subsection describing how to program a freshly recompiled bitstream using the Lattice Radiant Programmer; - reorganized the existing programming procedure under the Precompiled Bitstream subsection; - added the following note to the Precompiled Bitstream section, <i>Note: To open an existing programmer project, select Open an existing programmer project and browse to the project file (.xcf);</i> - updated Figure 6.6. Creating a New Project from a Scan, Figure 6.9. Device Properties Window for the CrossLinkU-NX SPI Flash Programming, and Figure 6.11. Programming Output Window. - In the Hardware Setup section, changed <i>Raspberry Pi Camera Module 2</i> to <i>Raspberry Pi Camera Module V2</i>. - In the Demonstration section, added the following note, <i>If no display appears after programming the precompiled or recompiled bitstream or display turns black when switching resolution during active display, refer to the Debugging section for detailed debugging instructions.</i>
Customizing the Reference Design	- In Table 7.1. IP Configuration for Supported Resolutions: - changed the resolution labels from <i>1080p</i> and <i>720p</i> to <i>1080p30</i> and <i>720p30</i>, respectively; - changed <i>clock frequency</i> to <i>Pixel clock's clock frequency</i> in Table Note 4. - In the Change the Camera Resolution section, added the following paragraph: <i>When a resolution change is requested, the host sends the resolution configuration to the on-board embedded firmware over USB using the UVC protocol. The firmware performs sensor reconfiguration only when the camera is idle and no active video stream is running, ensuring stable and uninterrupted operation.</i> - In the Selecting the 720p Output Resolution section, changed the Format drop-down selection from <i>YUV2 1920x1080</i> to <i>YUV2 1280x720</i> in Step 3. - In the Changing IP Configuration to 1 PPC section, added the following note: <i>The constraints file (.pdc) remains unchanged for this configuration.</i> - In the Configuring the MIPI RX IP section, added Step 5: <i>The unified video streaming data width has been updated, which causes existing UVSI connections to be disconnected. The connections can be re-established when isp_debayer_inst is reconfigured to the same PPC.</i> - In the Configuring the Debayer IP section, added Step 5: <i>The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect isp_debayer_inst AXI4S_S0 to mipi_csi_rx_inst AXI4S_VID as shown in Figure 7.9.</i> - In the Configuring the Color Correction Matrix IP section, added Step 5: <i>The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect isp_ccm_inst AXI4S_S0 to isp_debayer_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with mipi_csi_rx_inst through UVSI.</i> - In the Configuring the Color Space Converter IP section, added Step 5: <i>The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect isp_csc_inst AXI4S_S0 to isp_ccm_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with mipi_csi_rx_inst through UVSI.</i> - In the Configuring the Chroma Resampler IP section, added Step 5: <i>The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect isp_chroma_resampler_inst AXI4S_S0 to isp_csc_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with</i>

Section	Change Summary
	<i>mipi_csi_rx_inst</i> through UVSI. - In the Configuring the Video to USB Bridge section, added Step 5: <i>The unified video streaming data width has been updated, causing existing UVSI connections to be disconnected. Reconnect unified_video_to_usb_inst AXI4S_S0 to isp_chroma_resampler_inst AXI4S_M0. This is the same procedure used to connect isp_debayer_inst with mipi_csi_rx_inst via UVSI.</i> - Added Figure 7.9. Propel Builder Connection Update for UVSI.
Resource Utilization	Noted that the validated example configurations use <i>Lattice Radiant software 2025.2</i> .
Debugging	- Added the No Display After Programming the Bitstream section. - Added the Black Screen When Switching Resolution section. - Added the INFO and WARNING Messages In Propel Builder section.

Revision 1.2, March 2026

Section	Change Summary
All	Updated the reference design name from <i>CrossLinkU-NX USB Video Class</i> to <i>MIPI CSI-2 to USB UVC Reference Design</i> globally.
Introduction	- Updated Table 1.1. Summary of the Reference Design. - In the Features section, updated key features of the reference design to the current. - Updated the Naming Conventions section.
Directory Structure and Files	Updated Figure 2.1. Directory Structure.
Functional Description	- Updated Figure 3.1. Reference Design Block Diagram. - In the Design Components section: - removed the old PLL, MIPI Packet Decoder, Byte to Pixel, B2P to AXI Stream Conversion, AXI Stream to Parallel Conversion, YUV422 to UVC Bridge, IEBM in FIFO Interface, In Endpoint Buffer Manager, USB23 to AXI Bridge, RAW10 Test Pattern Generator, YUV422 Test Pattern In FIFO Bridge sections; - removed the old Figure 3.4. RISC-V MC IP Configuration and related description in the RISC-V Microcontroller section; - removed the old Figure 3.5. System Memory General Configuration, Figure 3.6. System Memory Port S0 Configuration, and Figure 3.7. System Memory Port S1 Configuration and related description in the System Memory section; - removed the old Figure 3.8. UART Configuration and related description in the UART section; - removed the old Figure 3.9. Hardware Version GPIO Configuration, Figure 3.10 Test Pattern Enable GPIO Configuration and related description in the GPIO section; - removed the old Figure 3.11 Lattice I2C Controller Configuration and updated the general description of the Lattice I2C Controller in the Lattice I2C Controller section; - removed the old Figure 3.16. APB Interconnect General Configuration and Figure 3.17. APB Interconnect Main Setting Configuration and related description in the APB Interconnect section; - removed the old Figure 3.18. APB-Lite to APB Converter Configuration and related description in the AHB-Lite to APB Converter section; - replaced the old RX DPHY section with the current MIPI CSI/DSI RX IP Core section; - replaced the old Debayer section with the current Debayer IP Core section; - removed the old Figure 3.25. Color Correction Matrix Configuration and updated the general description of the Color Correction Matrix IP Core in the Color Correction Matrix IP Core section; - removed the old Figure 3.26. Color Space Converter Input/Coefficient Configuration, Figure 3.27. Color Space Converter Output Configuration and updated general description of the Color Space Converter in the Color Space Converter IP Core section; - renamed the old AHB-Lite to Memory Bridge section to the current AHB-Lite to Native Memory Bridge section.

Section	Change Summary
	- added Chroma Resampler and Video to USB Bridge sections. - In the Clocking Scheme section, removed the old Figure 3.29. Reference Design Clock Domain Block Diagram and added Figure 3.2. Clocking Scheme Overview. - In the Reset Scheme section, removed the old Figure 3.30 Reference Design Reset Scheme Diagram and added Figure 3.3. Reset Scheme Overview.
In Endpoint Buffer Manager Architecture	Removed this section.
USB23 AXI Manager to Memory Interface Bridge	Removed this section.
Signal Description	Renamed the section name from <i>UVC Reference Design Signal Description</i> to the current and updated Table 4.1. Primary I/O.
Building the Reference Design	Removed this section.
Running the Reference Design	Added this section.
Running the Reference Design on Evaluation Board	Removed this section.
Implementing the Reference Design on Board	Added this section.
Customizing the Reference Design	Replaced the old Customizing the Reference Design section with detailed descriptions on the steps required to configure each IP and software for various demonstration setups.
Resource Utilization	Moved the resource utilization information to the current section and updated the data.
Debugging	Added this section.
References	Updated this section.

Revision 1.1, September 2025

Section	Change Summary
Functional Description	Updated Figure 3.1. Reference Design Block Diagram, Figure 3.10. Test Pattern Enable GPIO Configuration, Figure 3.29. Reference Design Clock Domain Block Diagram, and Figure 3.30. Reference Design Reset Scheme Diagram.
Building the Reference Design	- Updated Figure 7.12. Base Address Assignment. - In the Install IPs at Local from File section, added the following note: Note: for the latest version of the local IP files, contact Lattice Sales or Tech Support. - In the Opening Radiant Project section: - changed To create the FPGA bitstream file, open the Lattice Radiant software to To create the FPGA bitstream file, launch the Lattice Radiant software from Lattice Propel Builder; - changed The design should have no errors but there are 30 warnings due to some unused signals from some instance to The design should have no errors but there are 1074 warnings due to some unused signals from some instance; - updated Figure 7.15. Open the Radiant Design. - In the Generating Bitstream File section: - added If the Place & Route Design process encounters an out-of-memory error, reduce the number of host machine cores defined in Strategy1 > Place & Route Design from 7 to a smaller number; - added Figure 7.17. Place & Route Design Strategy.
LIFCL-33U Evaluation Board Programming	In the Radiant Programmer GUI Setup section, added <i>If the flash device is a GigaDevice flash, then the SPI Flash options are GD25LQ128E and 8-pin SOP. If the flash device is a Winbond flash, then the SPI Flash options are W25Q512JV and 8-pad WSON.</i>
Customizing the Reference Design	Added the Steps to Migrate from the FCCSP104 Package to the WLCSP84 Package section.

Revision 1.0, December 2024

Section	Change Summary
All	Production release.



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